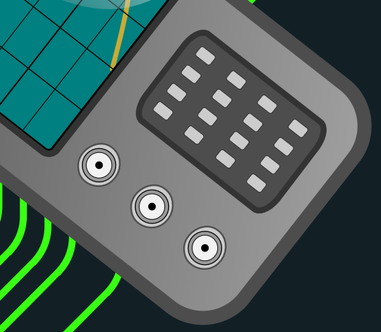
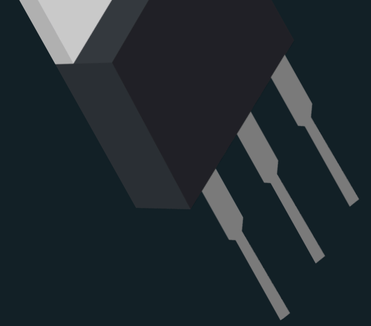
Single-Cycle MIPS Processor

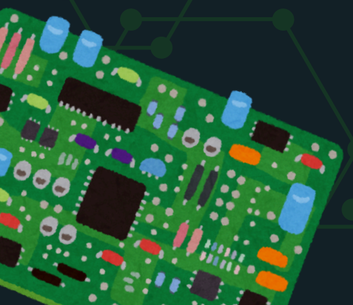
ECNG4104 Project Phase I

Our Team

Gamal Mohamed
Malak Mekky
Malak Abdelhalim







Contributions

Malak Abdelhalim

Register file + TB
Immediate extender + TB
Main control + TB

Processor Integration

Report + Presentation

Malak Mekky

ALU + TB
ALU control + TB

Processor Integration

FPGA Implementation

Gamal Mohamed

PC register + TB
Instruction memory + TB
Next-PC logic + TB

Processor Integration

FPGA Implementation

Specification Summary

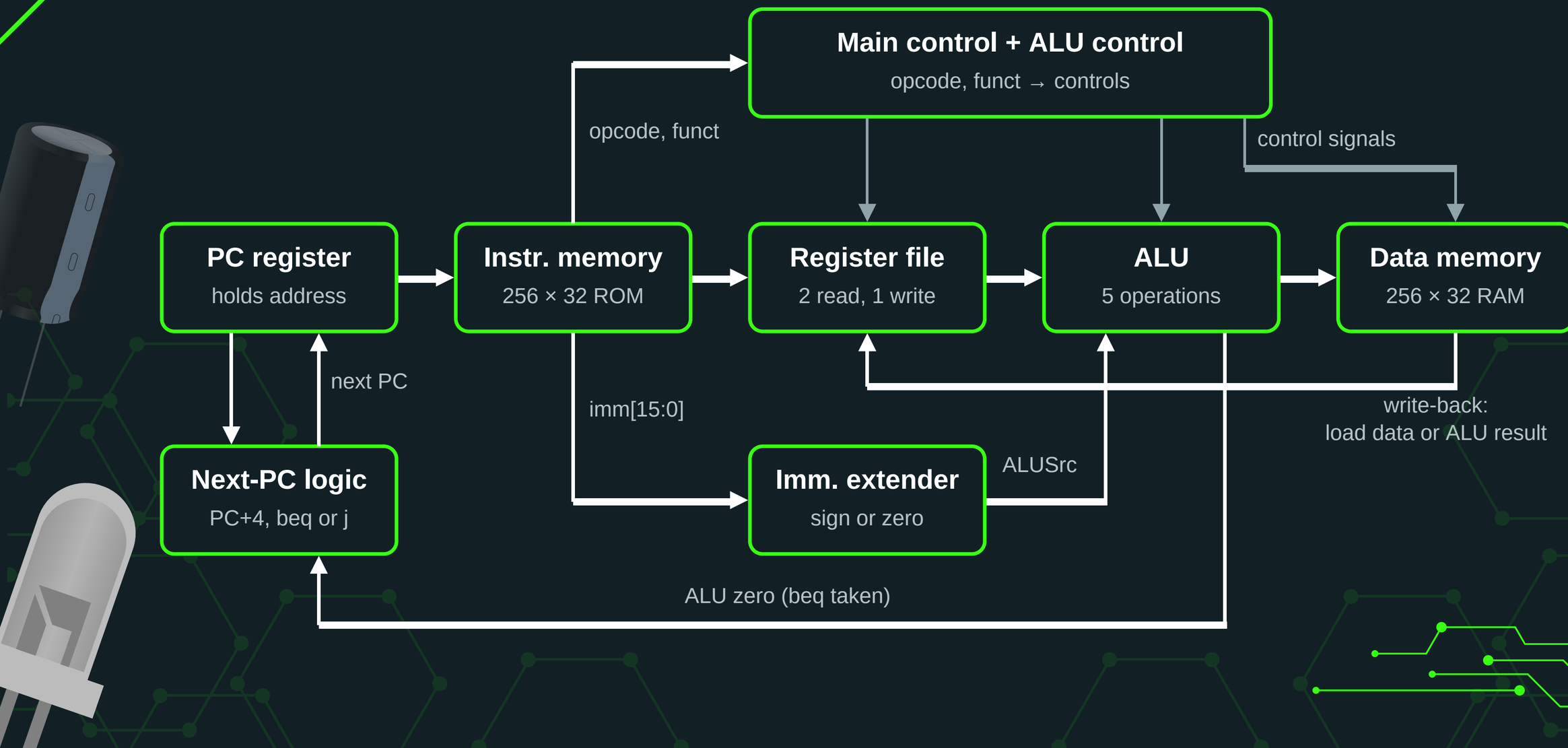
A 32-bit single-cycle MIPS processor. Every instruction is fetched, decoded, executed and written back in one clock cycle.

Instructions: add, sub, and, or, slt, addi, andi, lw, sw, beq, j

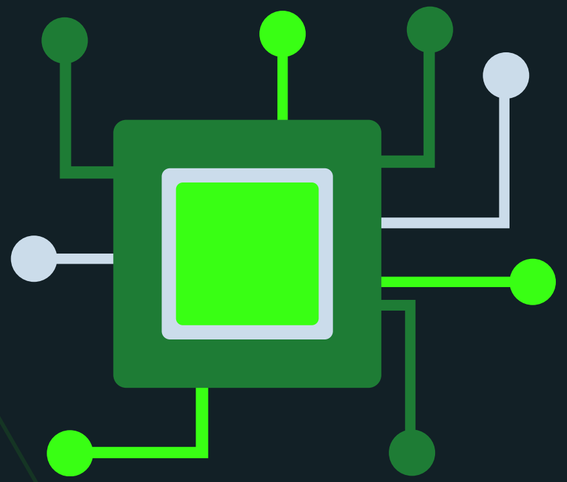
Phase I includes RTL design, module verification, processor integration, synthesis, place and route, and FPGA demonstration.

32-bit datapath
11 instructions
32 × 32 registers
256 × 32 memories
1 instruction per cycle
50 MHz board clock

Datapath & Control



Design Decisions



- R0 hard-wired to 0: reads return 0, writes ignored
- Reset added to the register file, so every run starts from known values
- Main control sets safe defaults (all 0): no latches & unknown opcodes do nothing
- Jump has priority over branch in the next-PC logic
- Memories read without a clock, so lw finishes in one cycle
- Branch and jump targets computed in the top level

RTL Walkthrough (1/3)

PC Register

32-bit register

Loads the next PC on the rising clock edge

Async active-low reset to 0

Instruction Memory

256 × 32 ROM

Loaded from a .hex file

Read without a clock, word index = addr[9:2]

Register File

32 × 32, 2 read ports and 1 write port

Writes on the clock edge, reads without a clock

R0 always 0. Reset clears all

RTL Walkthrough (2/3)

Imm. Extender

16 bits to 32 bits

Sign-extend for addi, lw, sw, beq

Zero-extend for andi

Main Control

Opcode: 10 control signals

All outputs default to 0, so no latches

Unknown opcode does nothing

ALU Control

ALUOp + funct = 3-bit ALU code

Follows the ALU control table

Defaults to ADD

RTL Walkthrough (3/3)

ALU

AND, OR, ADD, SUB and signed SLT

Zero flag drives beq

Unused codes give 0

Data Memory

256 × 32 RAM

Writes on the clock edge

Read without a clock, gated by MemRead

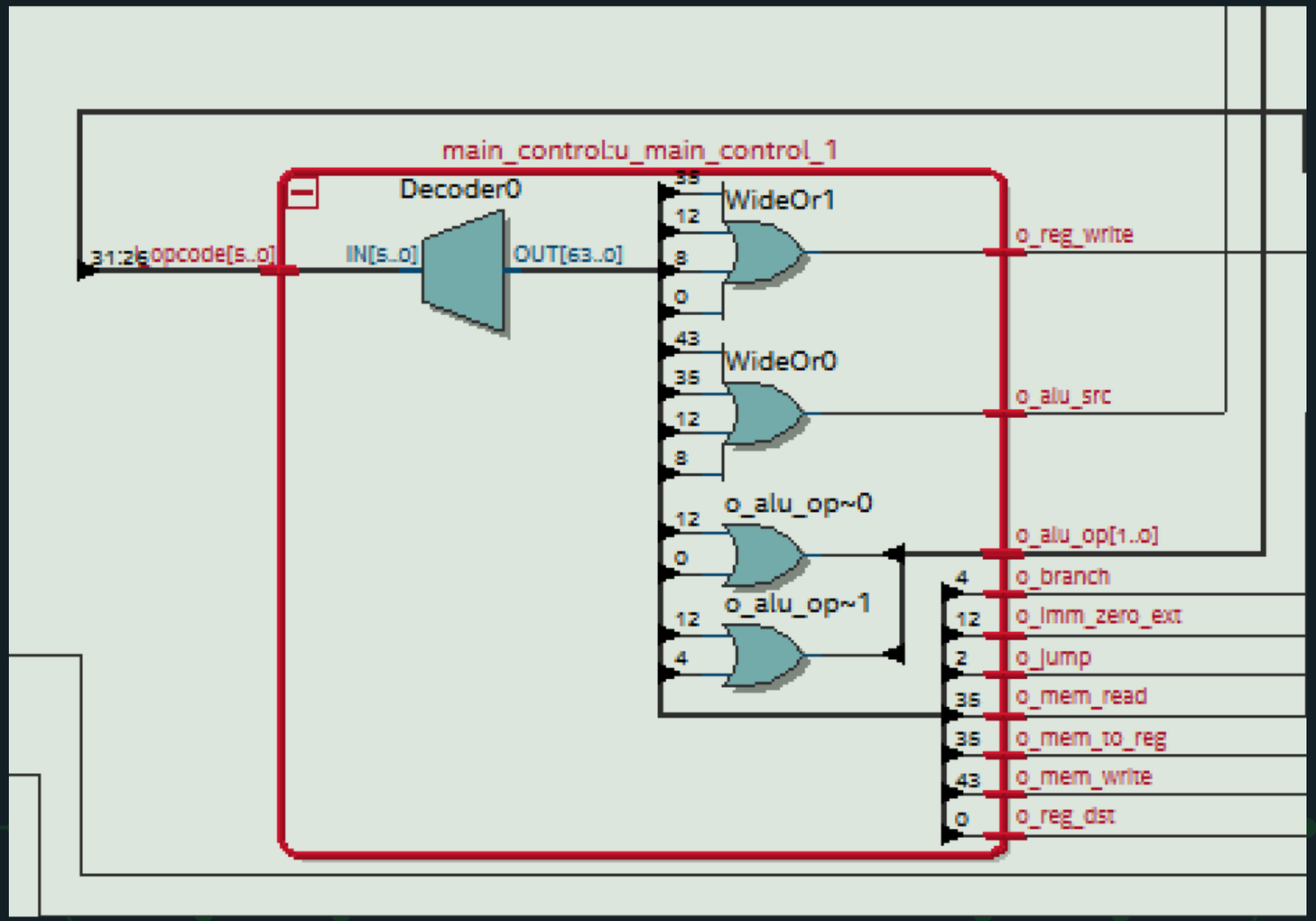
Next-PC Logic

PC + 4 by default

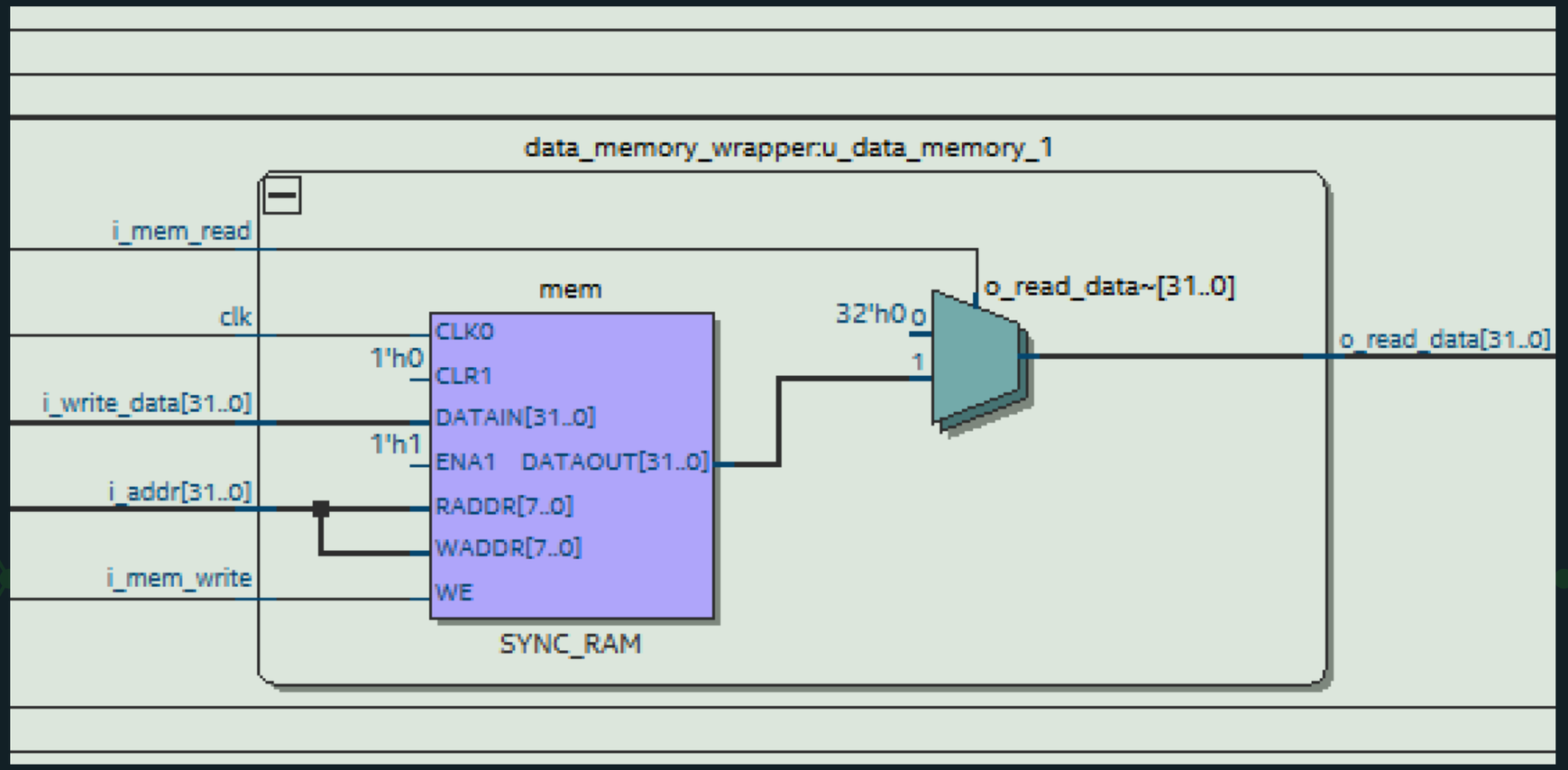
Branch target when beq is taken

Jump target has final priority

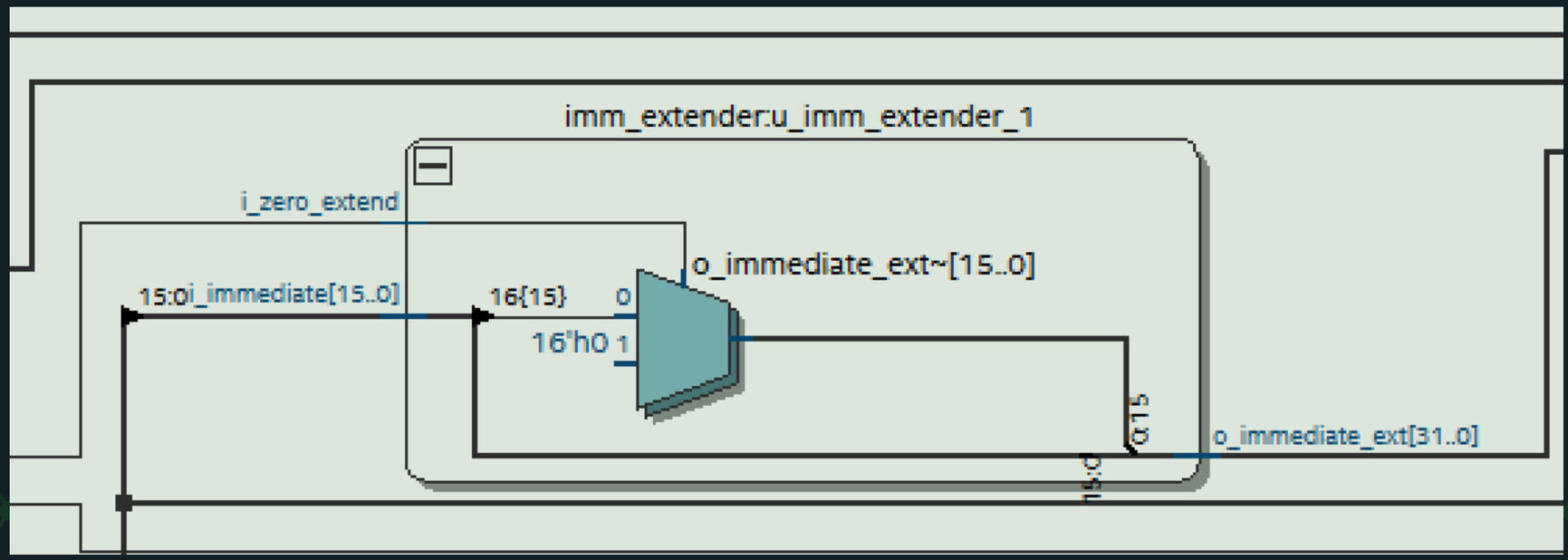
main control



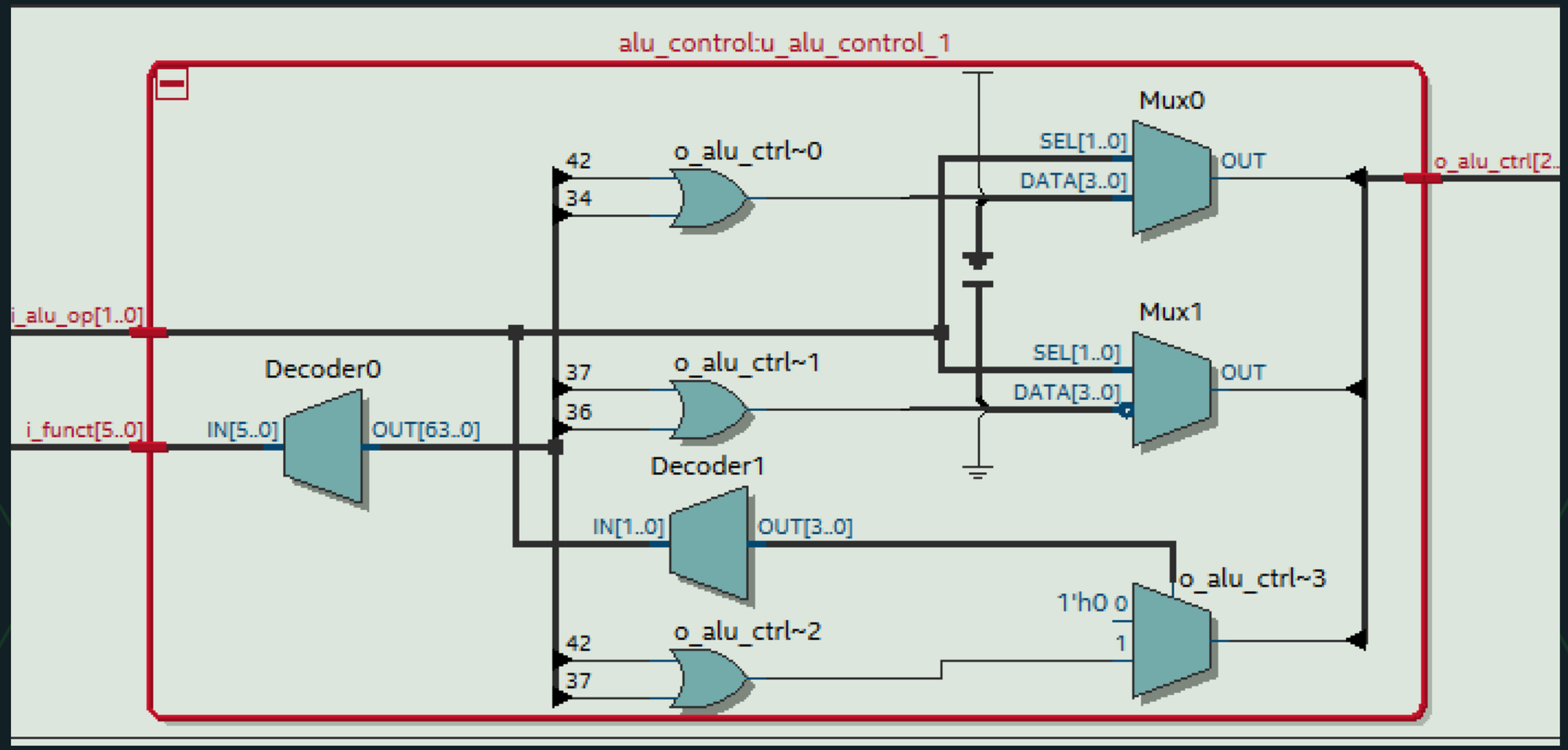
data memory wrapper



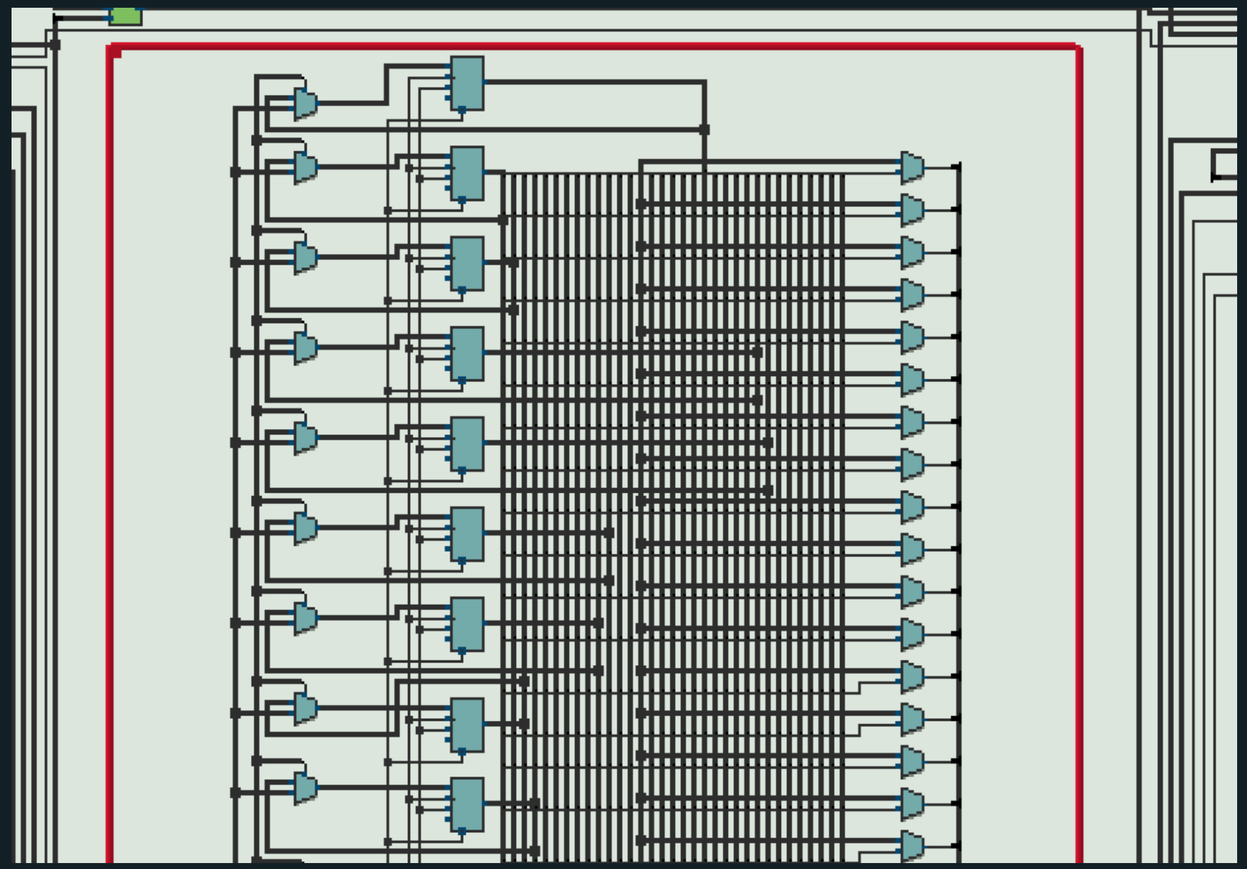
imm extender



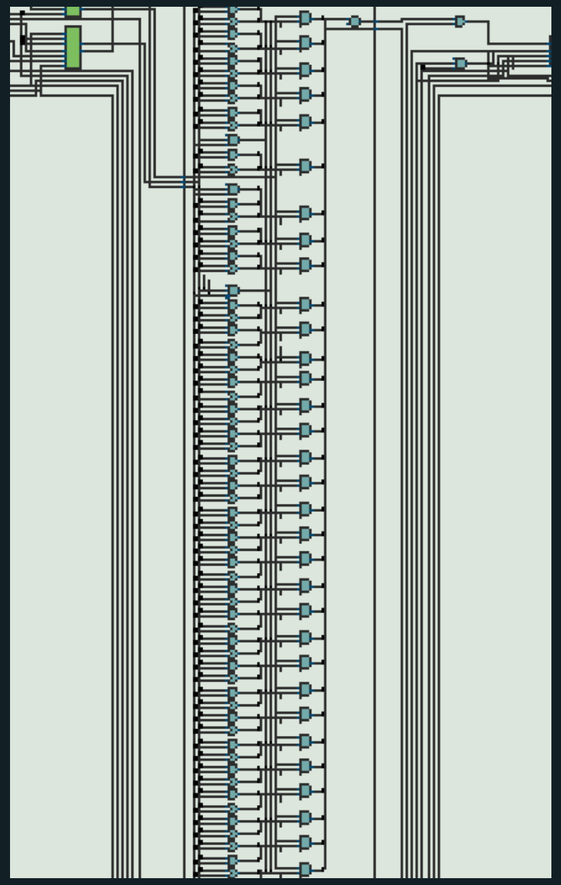
alu control


alu_control:u_alu_control_1
Mux0
SEL[1..0]
DATA[3..0]
OUT
o_alu_ctrl[2..
42
34
o_alu_ctrl~0
i_alu_op[1..0]
Mux1
SEL[1..0]
DATA[3..0]
OUT
Decoder0
37
36
o_alu_ctrl~1
i_funct[5..0]
IN[5..0]
OUT[63..0]
Decoder1
IN[1..0]
OUT[3..0]
o_alu_ctrl~3
1'h0 0
1
42
37
o_alu_ctrl~2

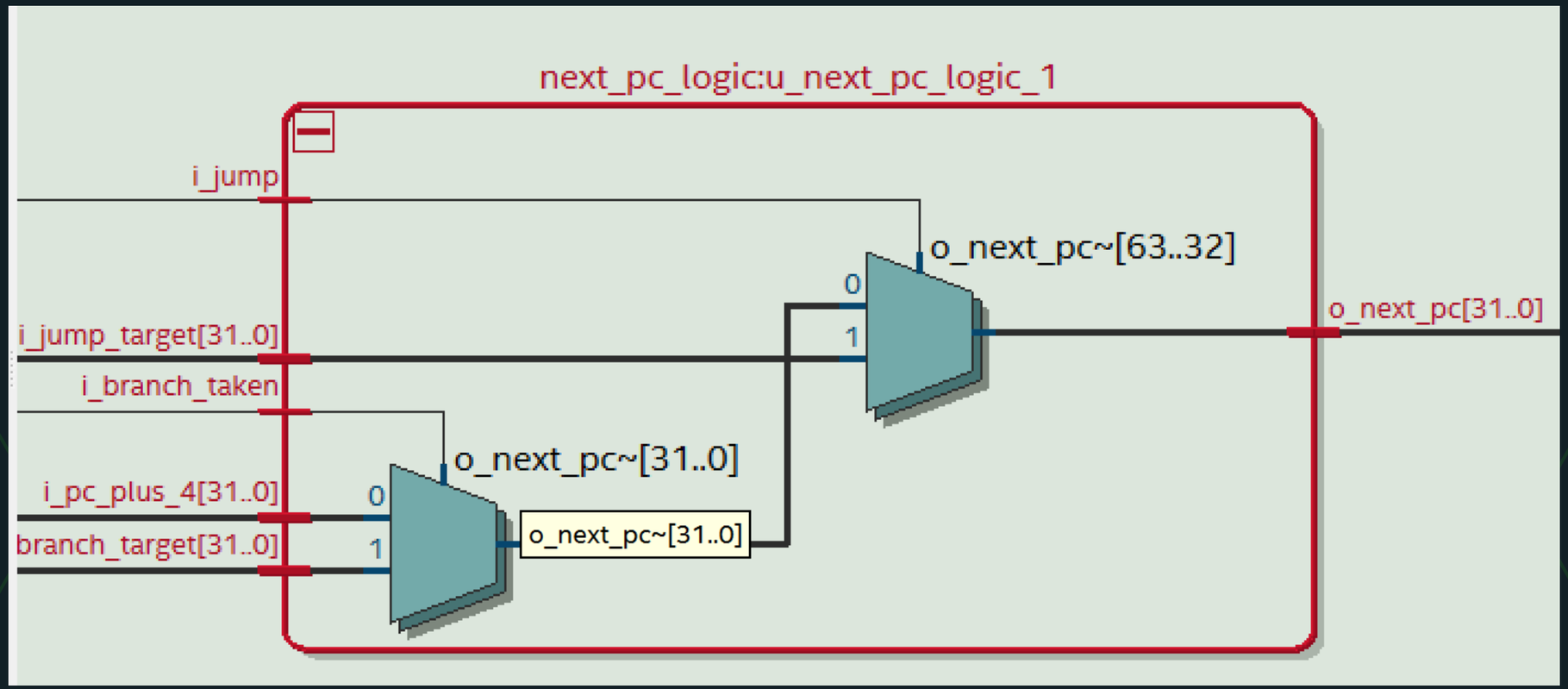
register file



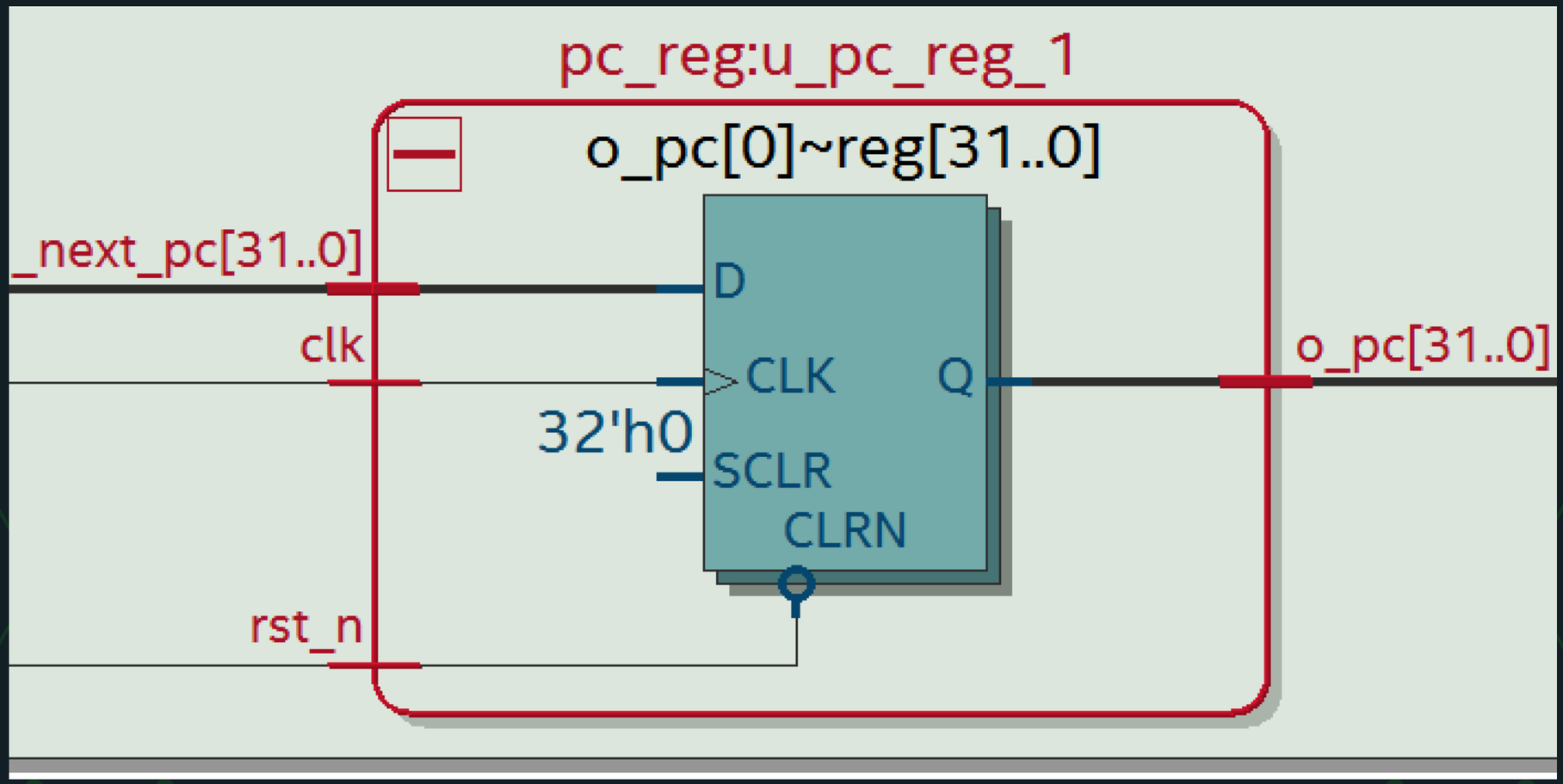
ALU



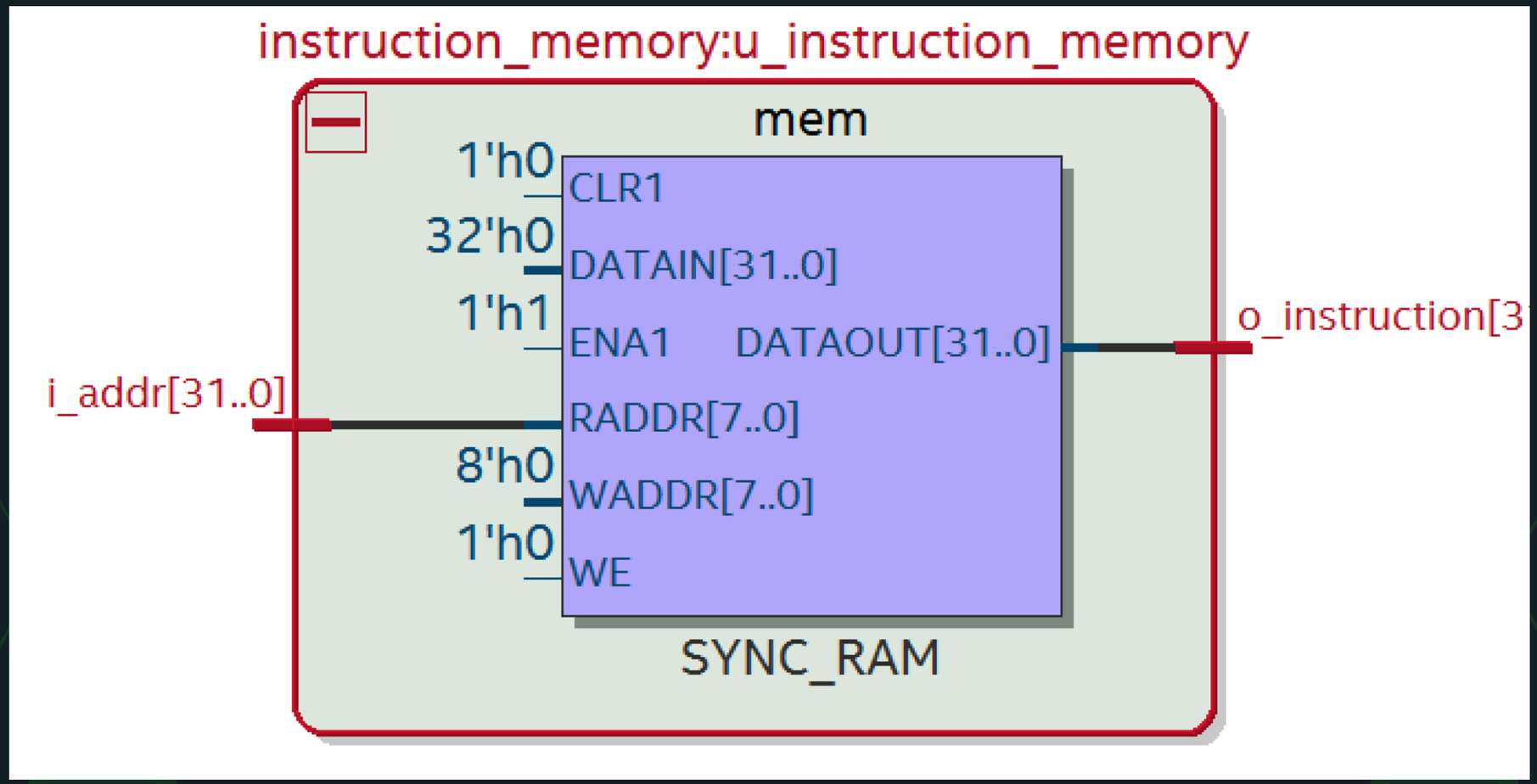
next PC logic



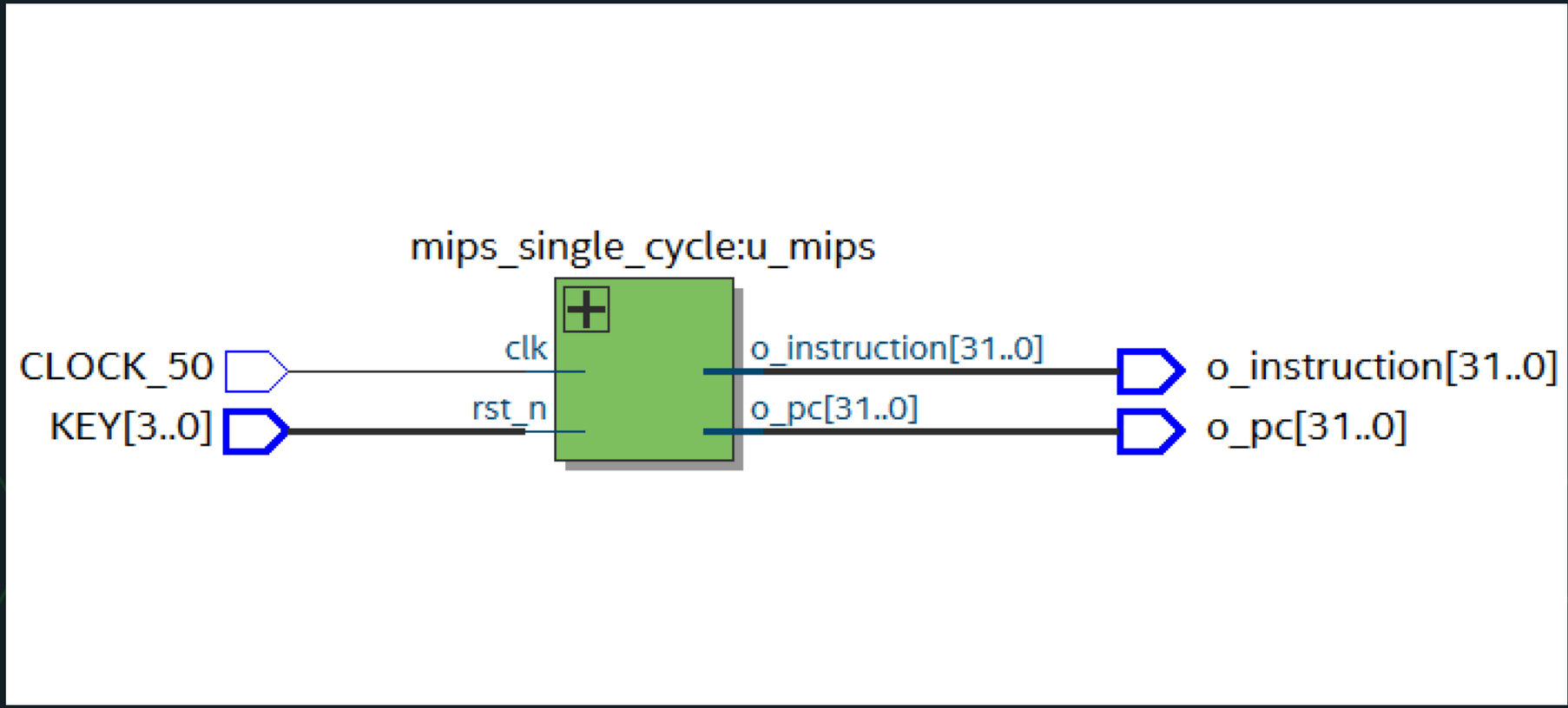
pc reg


pc_reg:u_pc_reg_1
o_pc[0]~reg[31..0]
_next_pc[31..0]
clk
32'h0
rst_n
D
CLK
Q
SCLR
CLRN
o_pc[31..0]

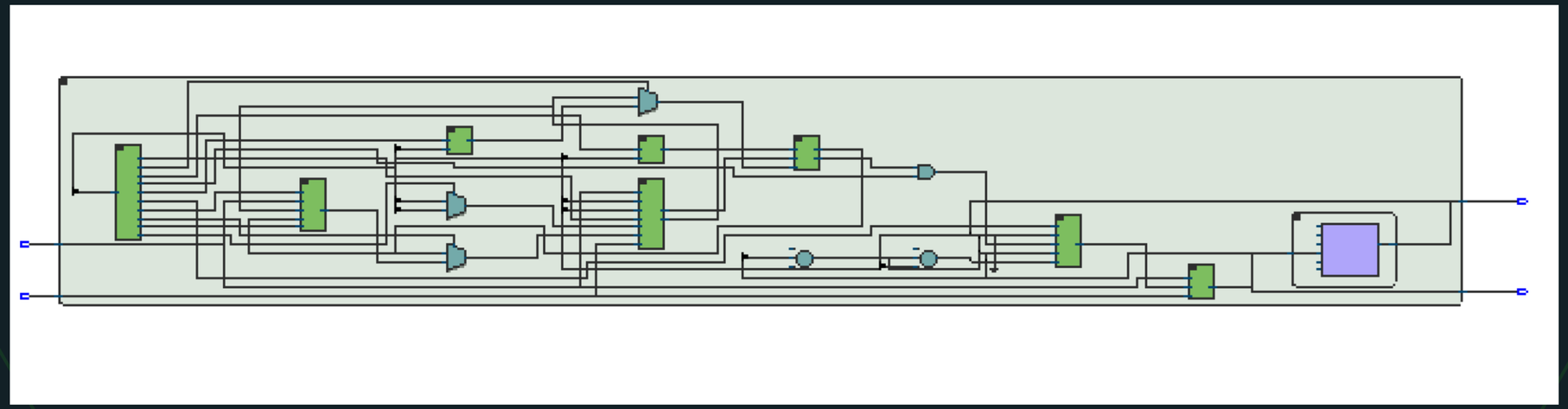
instruction memory

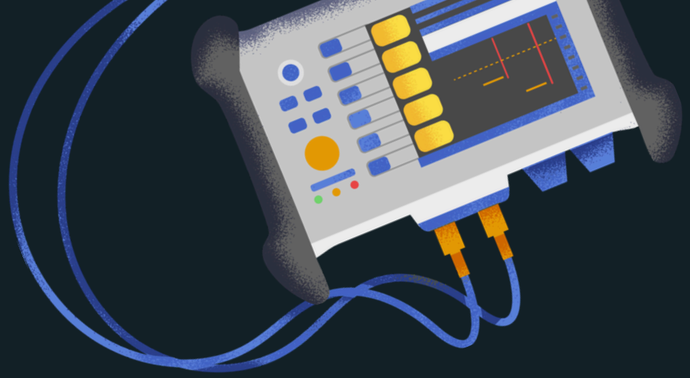


mips_single_cycle



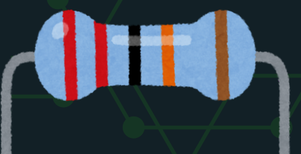
mips_single_cycle

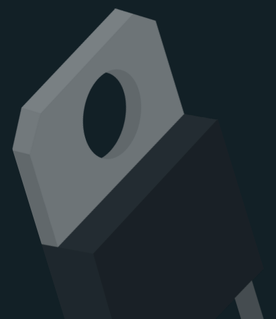




Verification Strategy

- Every module has a self-checking testbench: it compares outputs with expected values, counts errors, prints PASS or FAIL and stops
- Exhaustive if possible: all 64 opcodes, all 256 ALUOp × funct pairs
- ALU: 12 edge cases + 1,000 random pairs on all 8 control codes
- Processor: the four Appendix A programs, checking final registers, memory and PC





Test Programs

Test 1
Arithmetic & logic
R1 to R9 correct

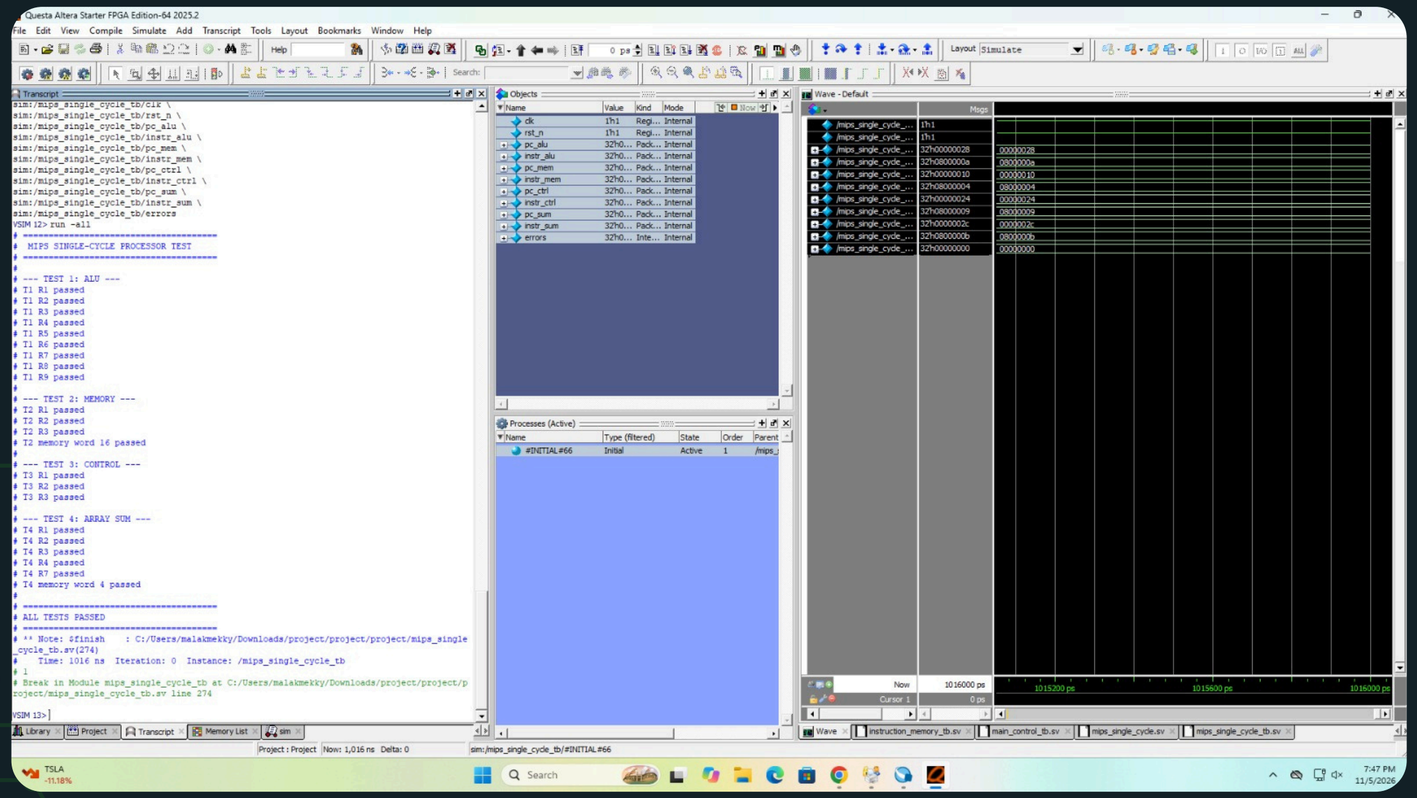
Test 2
Load & store
mem[16] = 42

Test 3
Branch & jump
R3 = 2

Test 4
Array sum
mem[4] = 25

All four programs pass in Questa. Tests 2, 3 and 4 were also checked on the DE1-SoC with SignalTap.

Processor Simulation

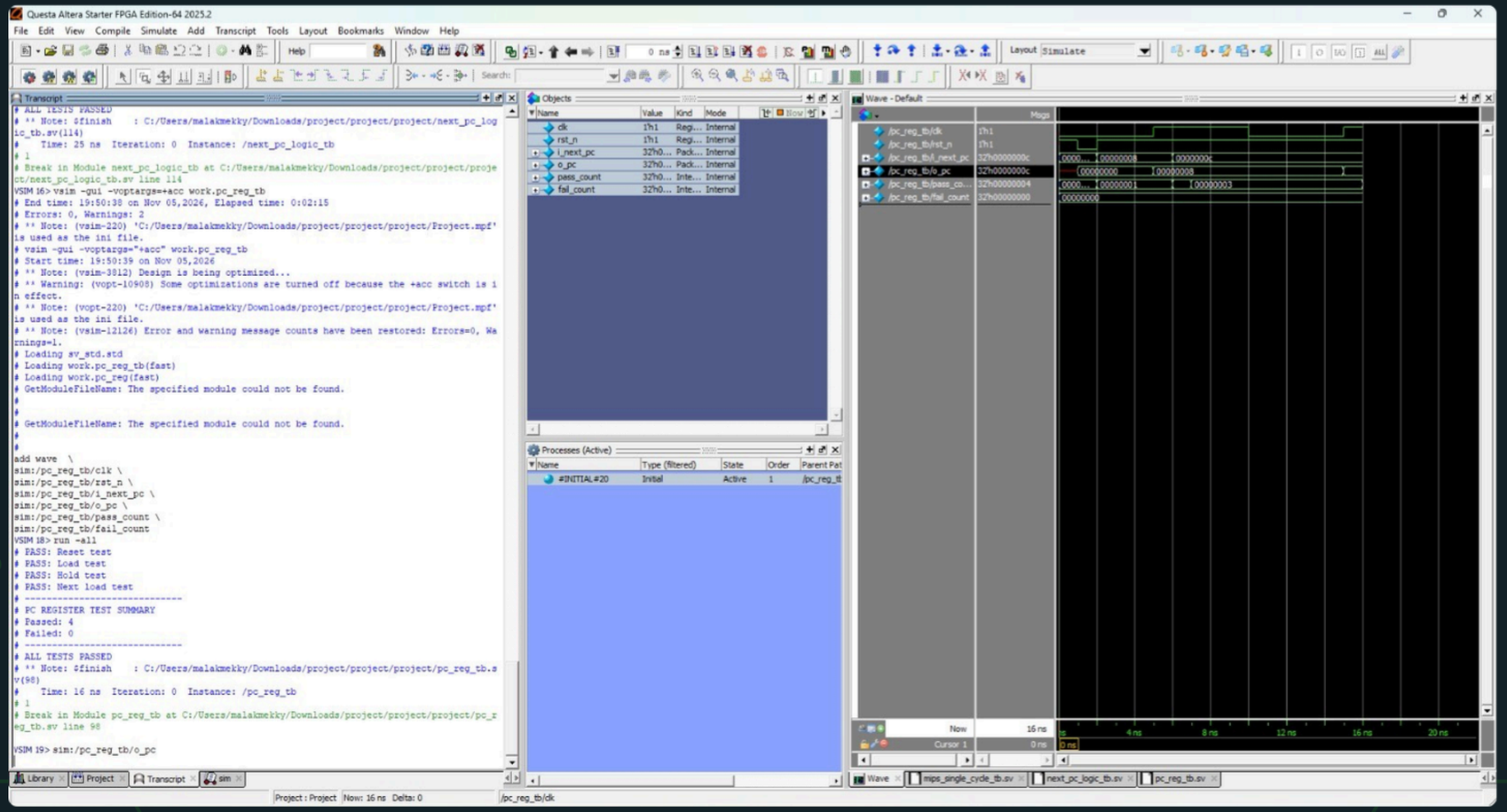


- One testbench runs all four programs at once, one processor per program
- Final PCs on the "j end" loops: 0x28, 0x10, 0x24, 0x2C
- R8 = 0xFFFFFFFF and R9 = 0x8001 confirm sign vs zero extension
- Result: ALL TESTS PASSED

Testbench Results

PC register	4/4	PASS
Register file	7/7	PASS
Imm. extender	8/8	PASS
ALU control	256/256	PASS
ALU	8,096/8,096	PASS
Main control	64/64	PASS
Data memory	6/6	PASS
Instr. memory	4/4	PASS
Next-PC logic	5/5	PASS
Processor	4/4 programs	PASS

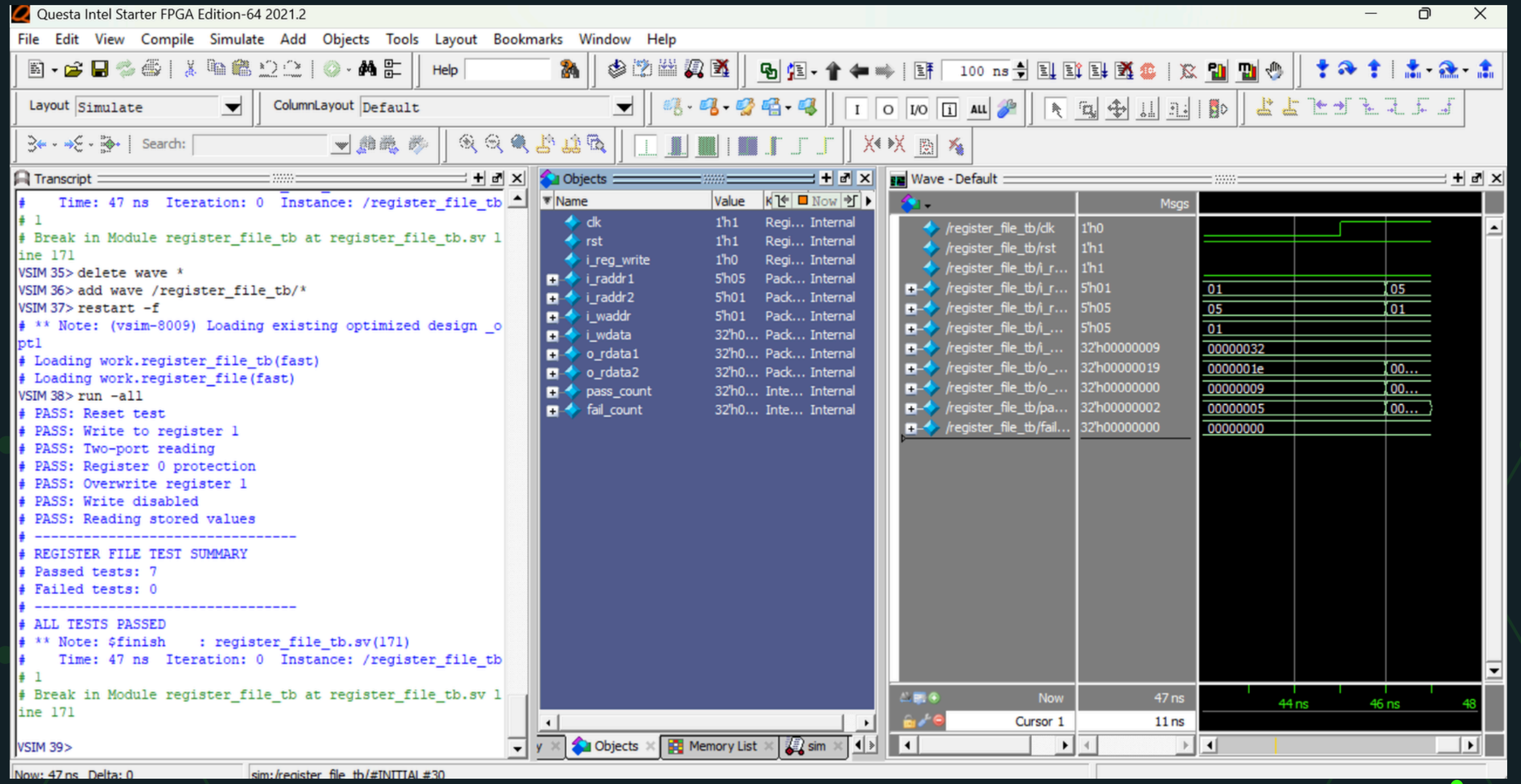
PC TB


Questa Altera Starter FPGA Edition-64 2025.2
File Edit View Compile Simulate Add Transcript Tools Layout Bookmarks Window Help
Transcript
ALL TESTS PASSED
** Note: $finish : C:/Users/malakmekky/Downloads/project/project/project/next_pc_logic_tb.sv(114)
Time: 25 ns Iteration: 0 Instance: /next_pc_logic_tb
1
Break in Module next_pc_logic_tb at C:/Users/malakmekky/Downloads/project/project/project/next_pc_logic_tb.sv line 114
VSIM 16> vsim -gui -voptargs=+acc work.pc_reg_tb
End time: 19:50:38 on Nov 05,2026, Elapsed time: 0:02:15
Errors: 0, Warnings: 2
** Note: (vsim-220) 'C:/Users/malakmekky/Downloads/project/project/project/Project.mpf' is used as the ini file.
vsim -gui -voptargs="+acc" work.pc_reg_tb
Start time: 19:50:39 on Nov 05,2026
** Note: (vsim-3812) Design is being optimized...
** Warning: (vopt-10908) Some optimizations are turned off because the +acc switch is in effect.
** Note: (vopt-220) 'C:/Users/malakmekky/Downloads/project/project/project/Project.mpf' is used as the ini file.
** Note: (vsim-12126) Error and warning message counts have been restored: Errors=0, Warnings=1.
Loading sv_std.std
Loading work.pc_reg_tb(fast)
Loading work.pc_reg(fast)
GetModuleFileName: The specified module could not be found.
#
#
GetModuleFileName: The specified module could not be found.
#
#
add wave \
sim:/pc_reg_tb/clk \
sim:/pc_reg_tb/rst_n \
sim:/pc_reg_tb/i_next_pc \
sim:/pc_reg_tb/o_pc \
sim:/pc_reg_tb/pass_count \
sim:/pc_reg_tb/fail_count
VSIM 18> run -all
PASS: Reset test
PASS: Load test
PASS: Hold test
PASS: Next load test

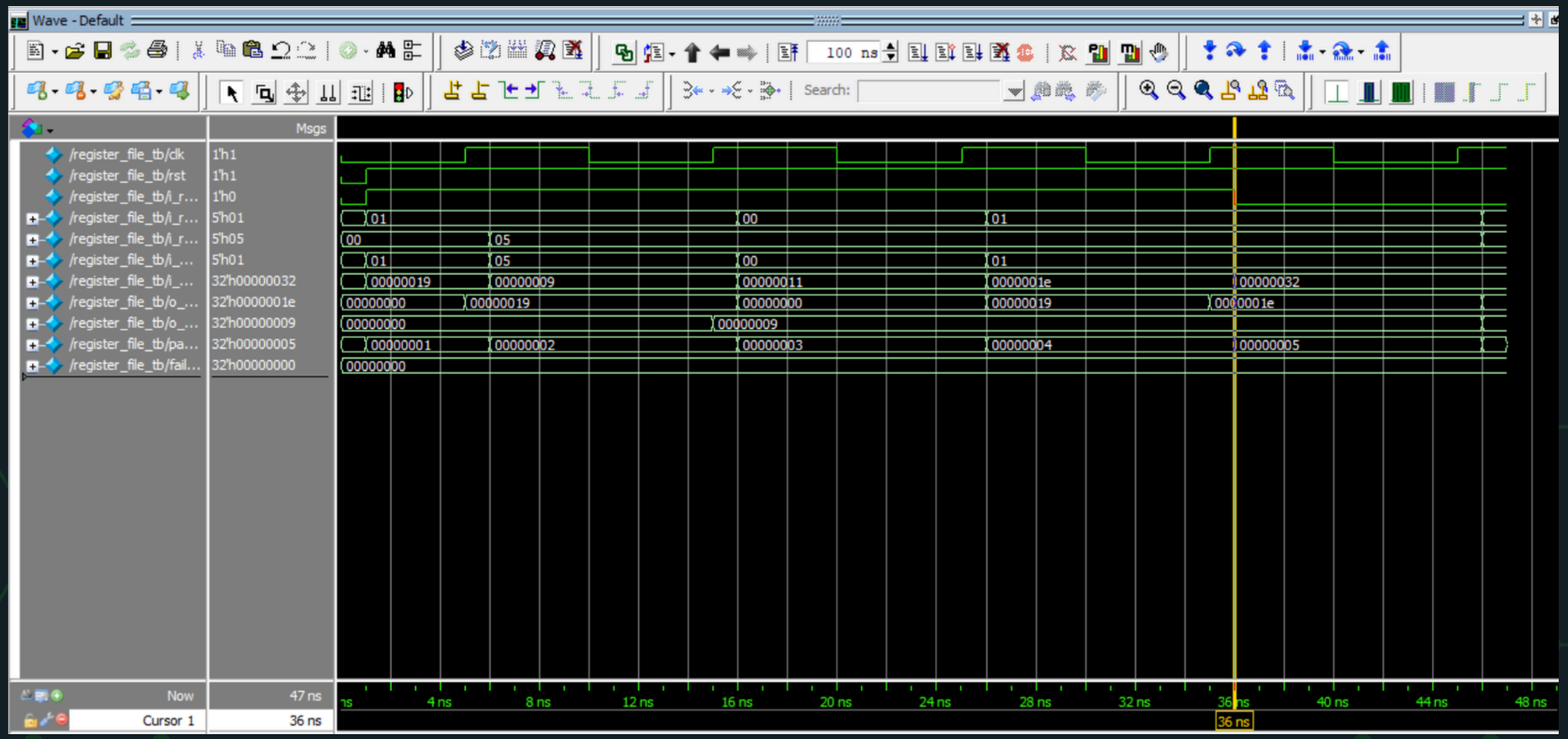
PC REGISTER TEST SUMMARY
Passed: 4
Failed: 0

ALL TESTS PASSED
** Note: $finish : C:/Users/malakmekky/Downloads/project/project/project/pc_reg_tb.sv(98)
Time: 16 ns Iteration: 0 Instance: /pc_reg_tb
1
Break in Module pc_reg_tb at C:/Users/malakmekky/Downloads/project/project/project/pc_reg_tb.sv line 98
VSIM 19> sim:/pc_reg_tb/o_pc
Objects
Name Value Kind Mode
clk 1'h1 Reg... Internal
rst_n 1'h1 Reg... Internal
i_next_pc 32'h0... Pack... Internal
o_pc 32'h0... Pack... Internal
pass_count 32'h0... Inte... Internal
fail_count 32'h0... Inte... Internal
Processes (Active)
Name Type (filtered) State Order Parent Pat
#INITIAL#20 Initial Active 1 /pc_reg_tb
Wave - Default
/pc_reg_tb/clk
/pc_reg_tb/rst_n
/pc_reg_tb/i_next_pc
/pc_reg_tb/o_pc
/pc_reg_tb/pass_co...
/pc_reg_tb/fail_count
Now 16 ns
Cursor 1 0 ns
Library Project Transcript sim
Wave mips_single_cycle_tb.sv next_pc_logic_tb.sv pc_reg_tb.sv
Project : Project Now: 16 ns Delta: 0
/pc_reg_tb/clk

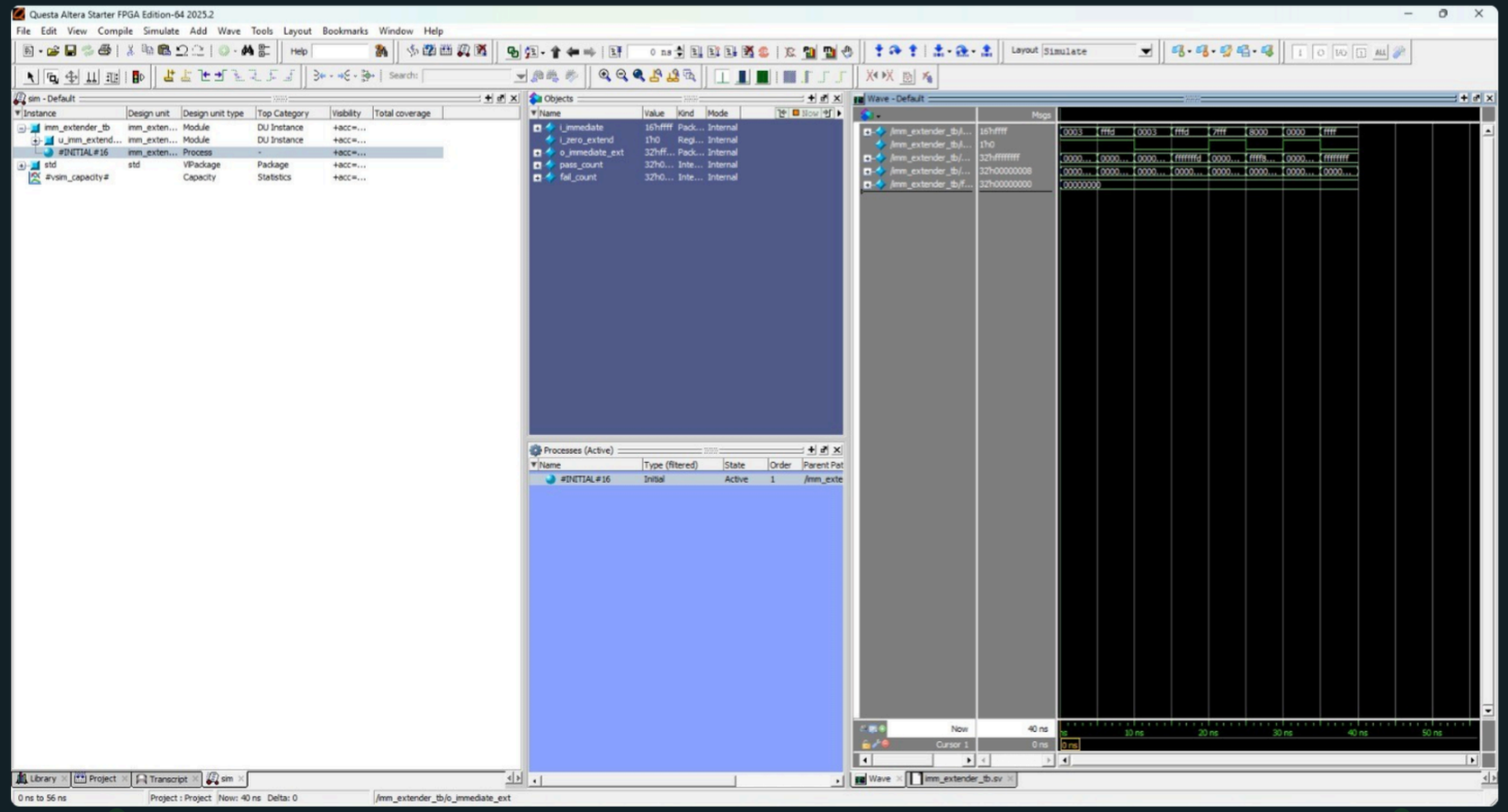
register file TB (1/2)



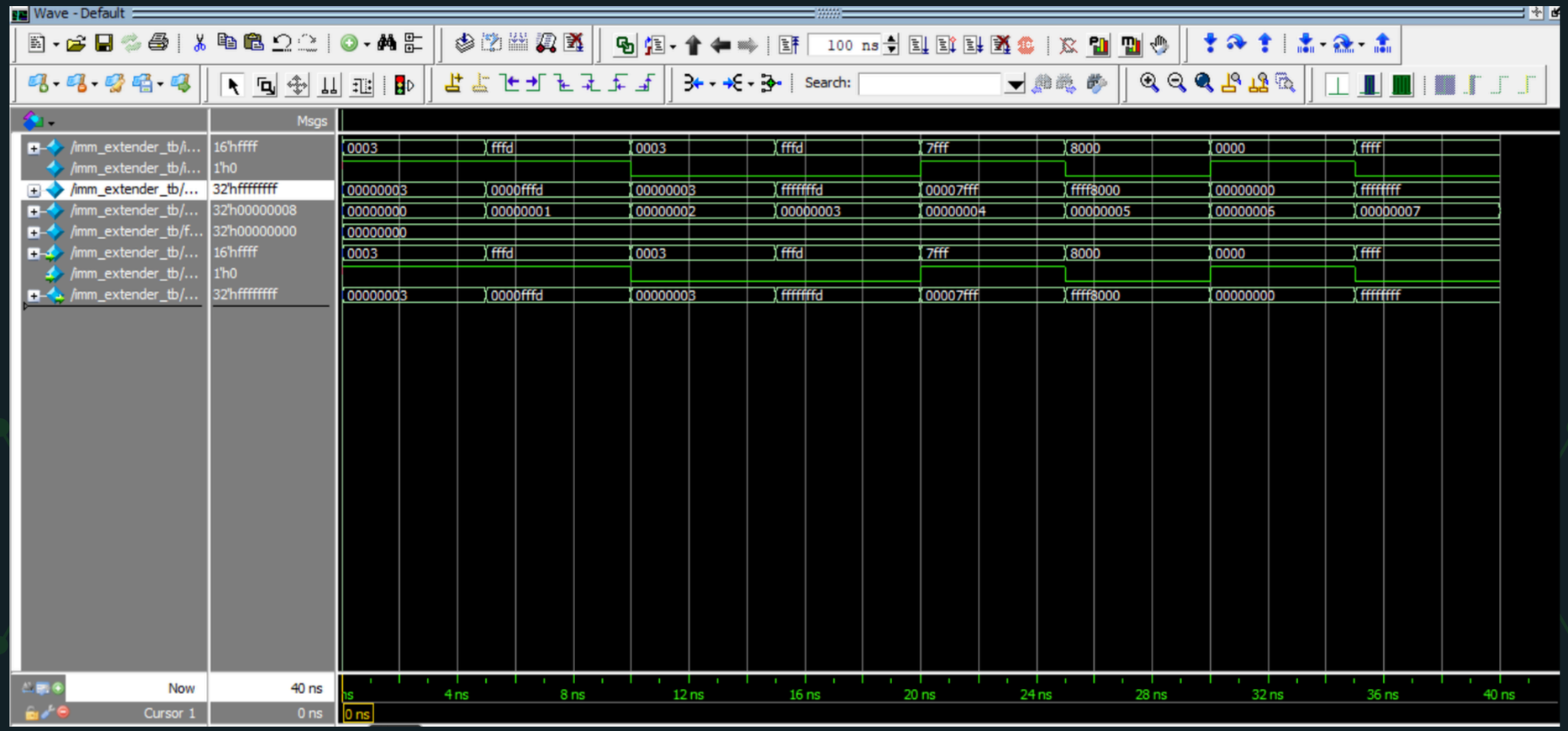
register file TB (2/2)



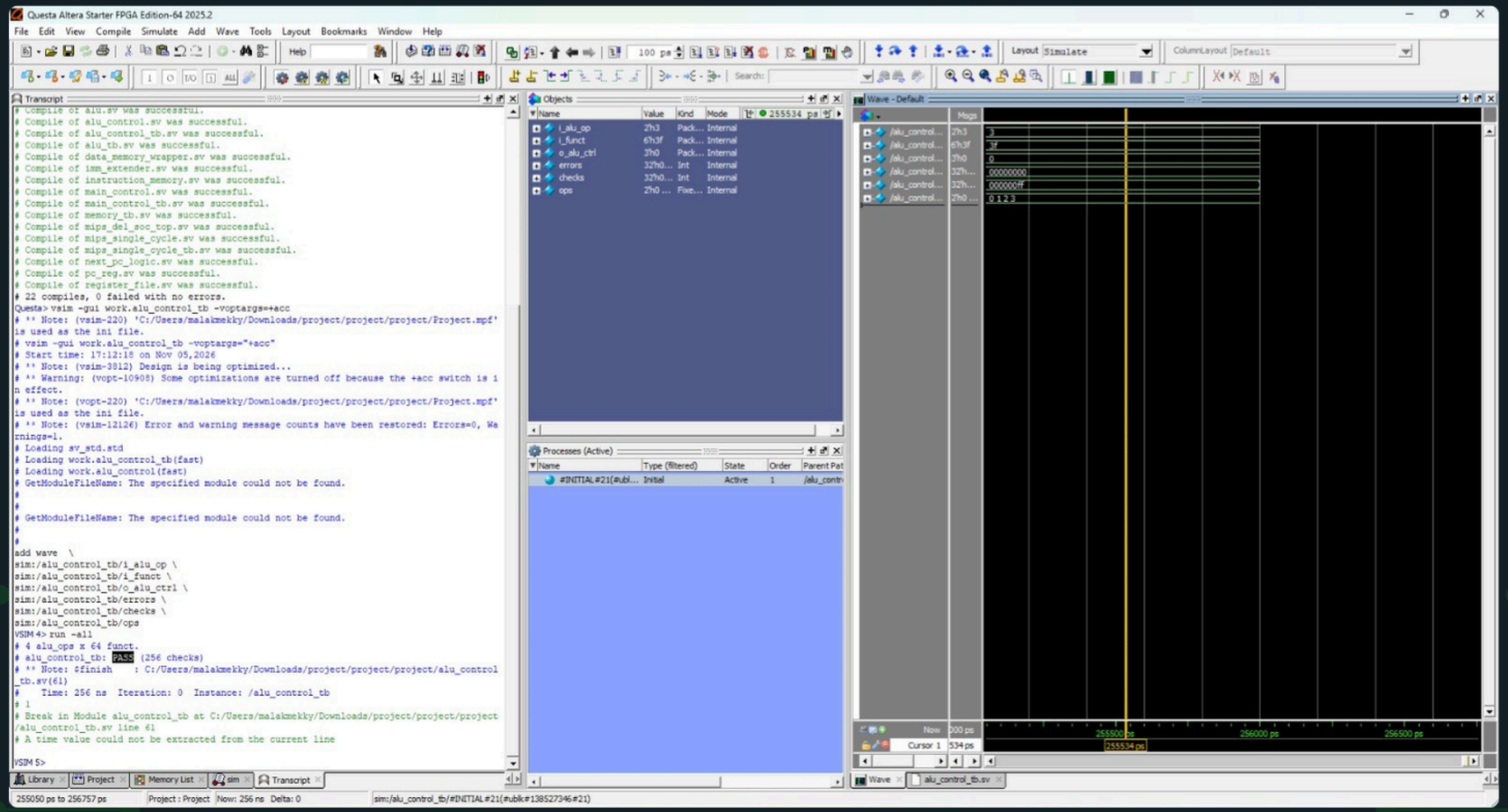
imm extender TB (1/2)



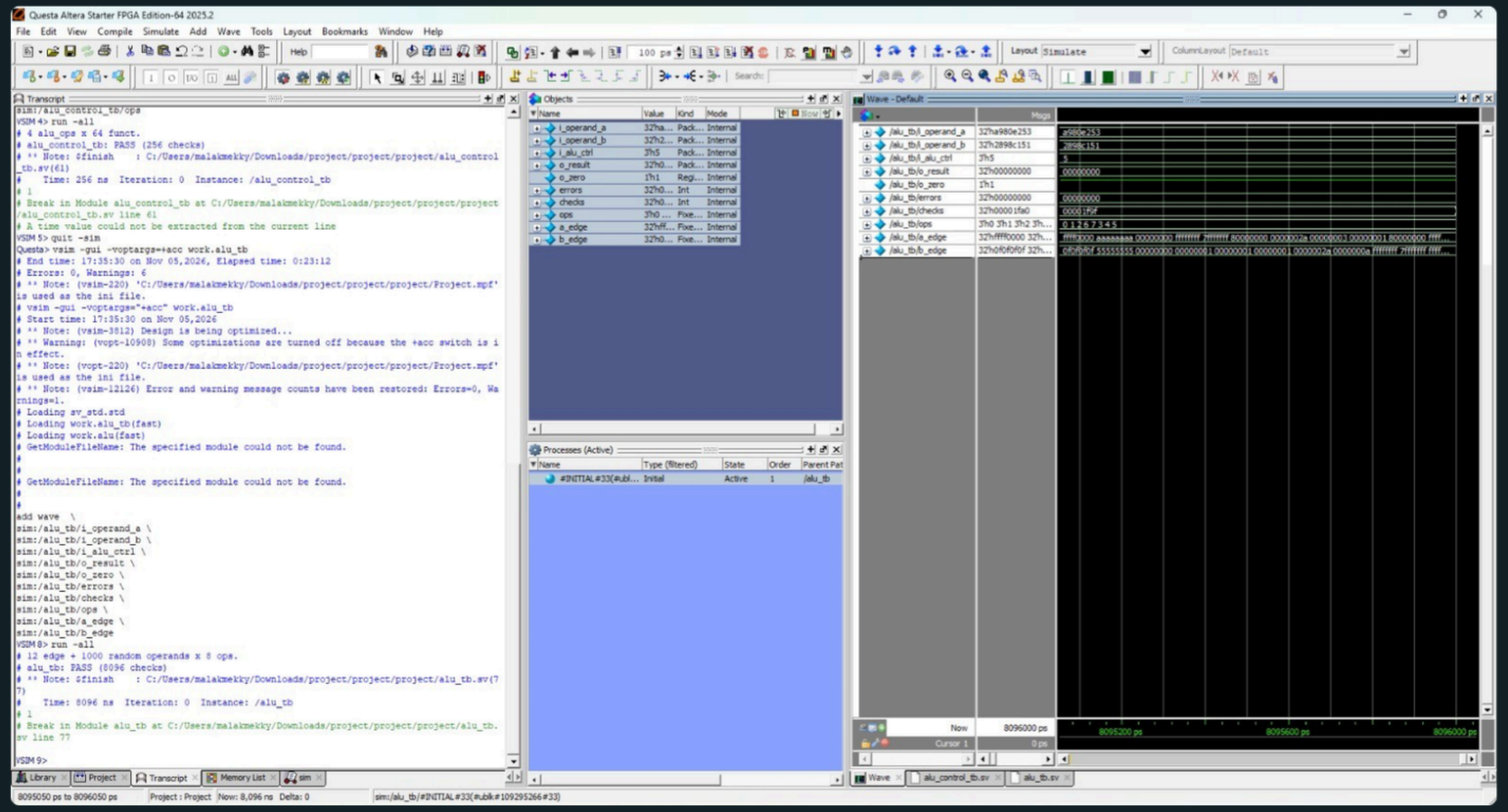
imm extender TB (2/2)



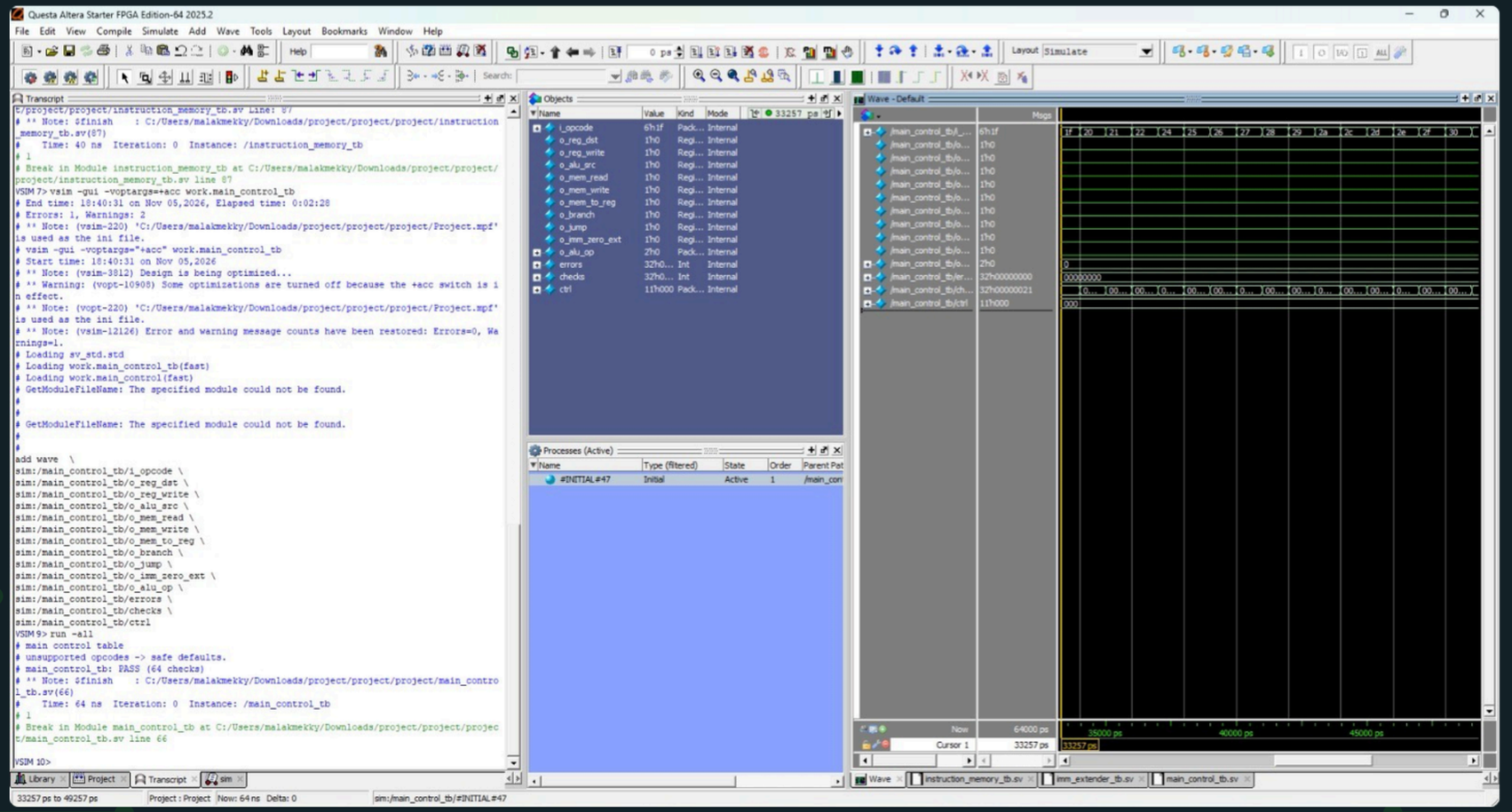
alu control TB



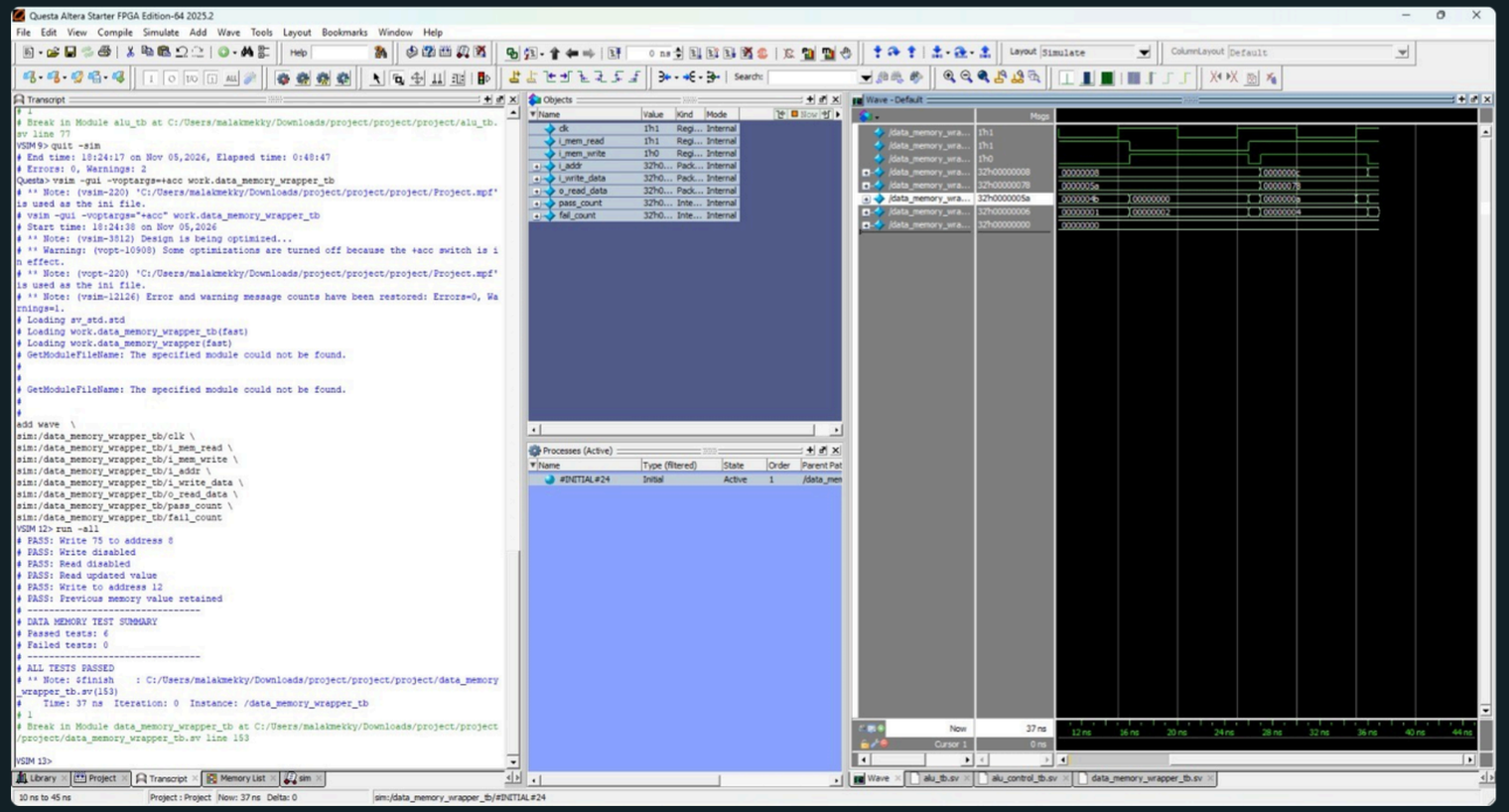
ALU TB


Questa Altera Starter FPGA Edition-64 2025.2
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
Transcript
sim:/alu_control_tb/ops
VSIM 4> run -all
4 alu_ops x 64 funct.
alu_control_tb: PASS (256 checks)
** Note: $finish : C:/Users/malakmekky/Downloads/project/project/project/alu_control_tb.sv(61)
Time: 256 ns Iteration: 0 Instance: /alu_control_tb
1
Break in Module alu_control_tb at C:/Users/malakmekky/Downloads/project/project/project/alu_control_tb.sv line 61
A time value could not be extracted from the current line
VSIM 5> quit -sim
Questa> vsim -gui -voptargs=+acc work.alu_tb
End time: 17:35:30 on Nov 05,2026, Elapsed time: 0:23:12
Errors: 0, Warnings: 6
** Note: (vsim-220) 'C:/Users/malakmekky/Downloads/project/project/project/Project.mpf' is used as the ini file.
vsim -gui -voptargs="+acc" work.alu_tb
Start time: 17:35:30 on Nov 05,2026
** Note: (vsim-3812) Design is being optimized...
** Warning: (vopt-10908) Some optimizations are turned off because the +acc switch is in effect.
** Note: (vopt-220) 'C:/Users/malakmekky/Downloads/project/project/project/Project.mpf' is used as the ini file.
** Note: (vsim-12126) Error and warning message counts have been restored: Errors=0, Warnings=1.
Loading sv_std.std
Loading work.alu_tb(fast)
Loading work.alu(fast)
GetModuleFileName: The specified module could not be found.
GetModuleFileName: The specified module could not be found.
add wave \
sim:/alu_tb/i_operand_a \
sim:/alu_tb/i_operand_b \
sim:/alu_tb/i_alu_ctrl \
sim:/alu_tb/o_result \
sim:/alu_tb/o_zero \
sim:/alu_tb/errors \
sim:/alu_tb/checks \
sim:/alu_tb/ops \
sim:/alu_tb/a_edge \
sim:/alu_tb/b_edge
VSIM 8> run -all
12 edge + 1000 random operands x 8 ops.
alu_tb: PASS (8096 checks)
** Note: $finish : C:/Users/malakmekky/Downloads/project/project/project/alu_tb.sv(77)
Time: 8096 ns Iteration: 0 Instance: /alu_tb
1
Break in Module alu_tb at C:/Users/malakmekky/Downloads/project/project/project/alu_tb.sv line 77
VSIM 9>
Objects
Processes (Active)
Wave - Default
Now
Cursor 1

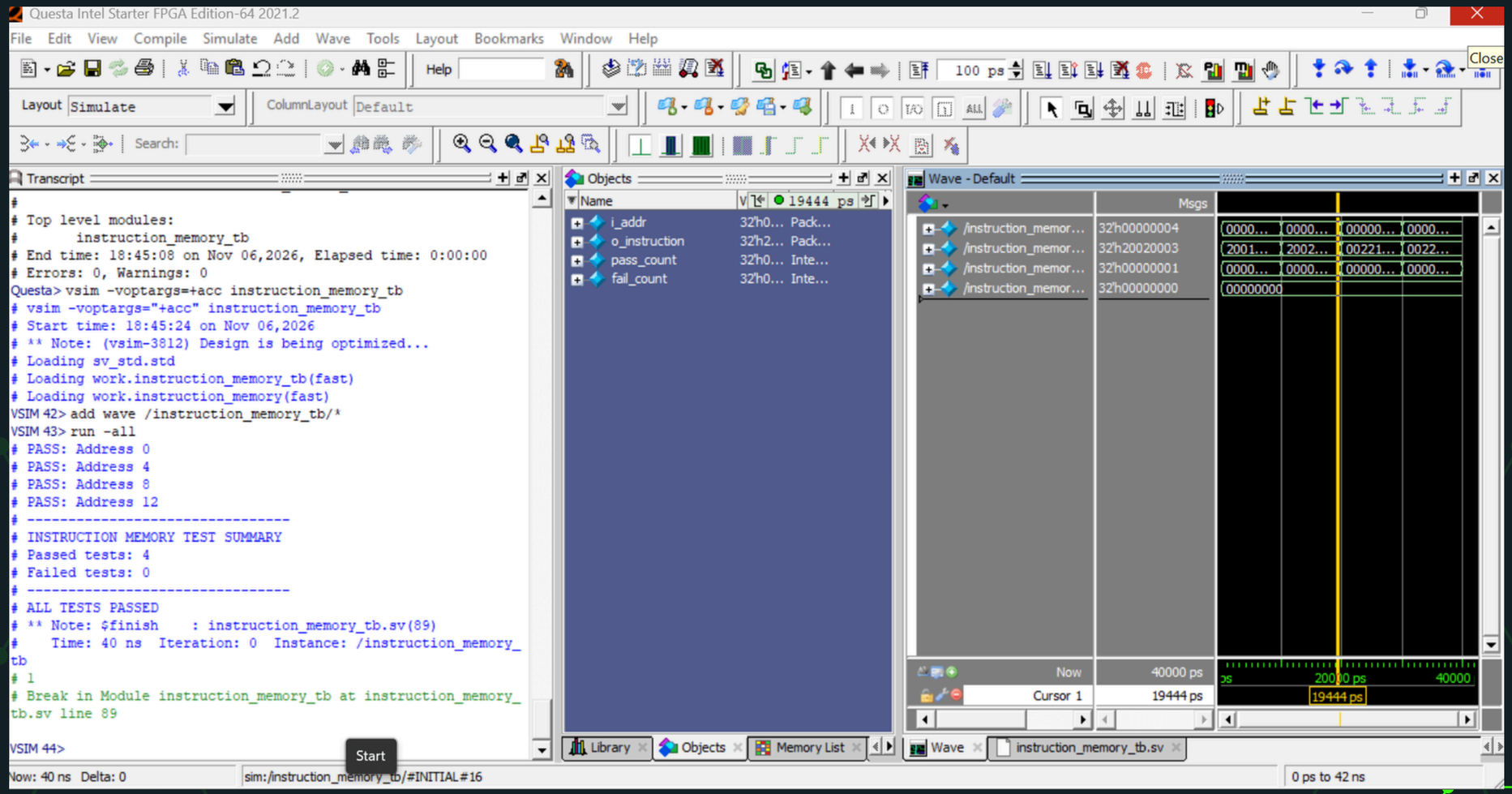
main control TB



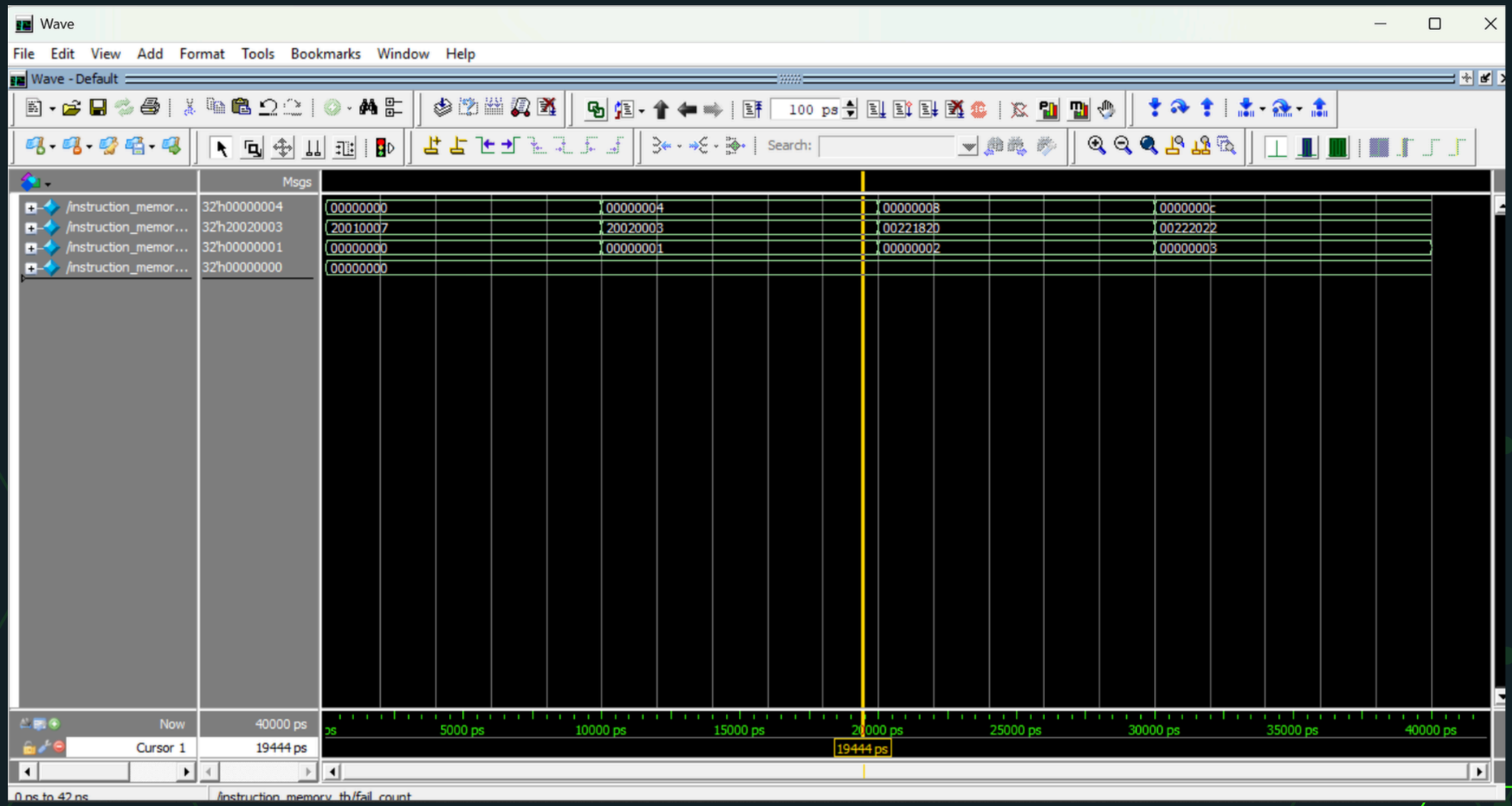
data memory TB



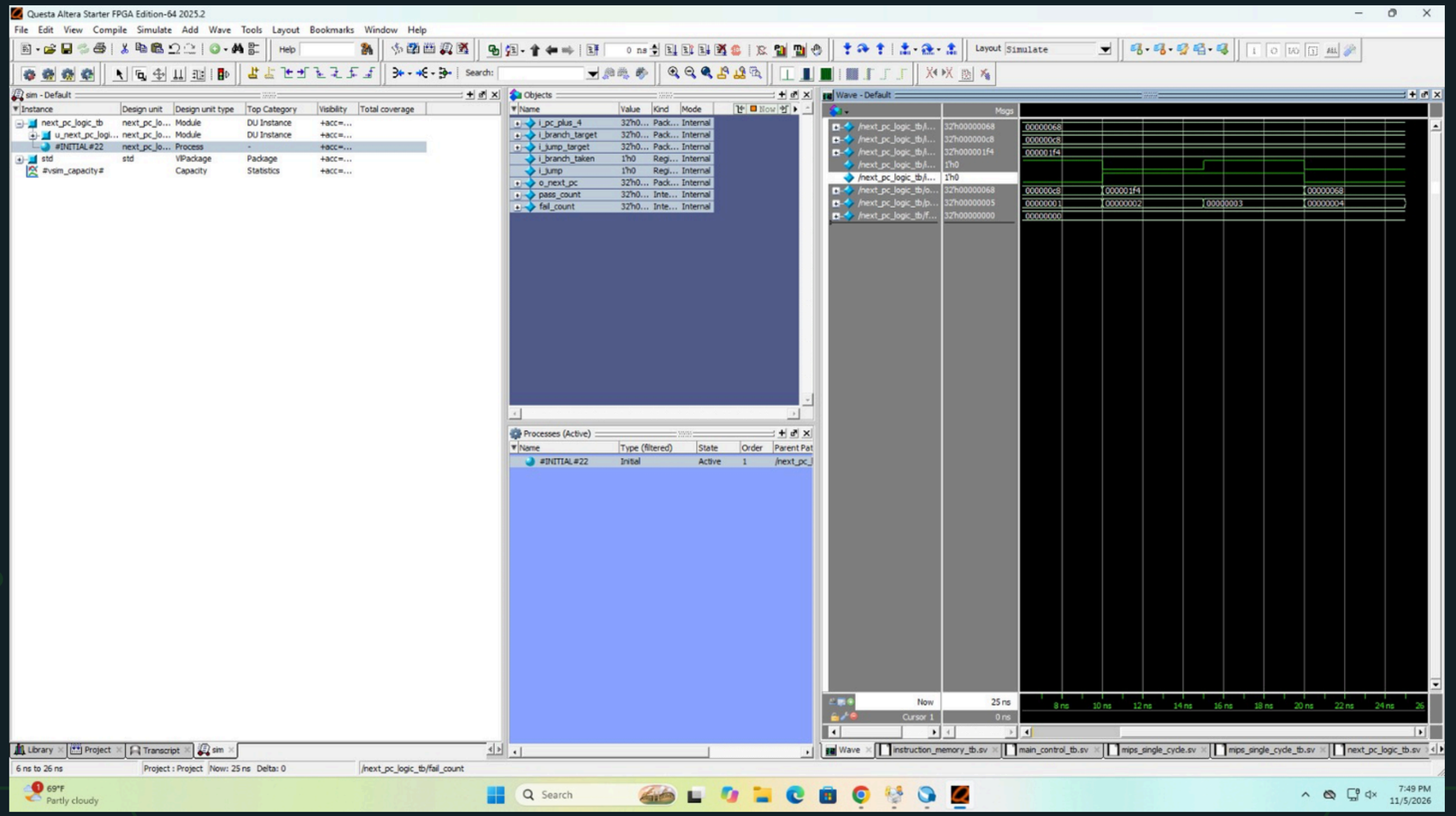
instruction memory TB (1/2)



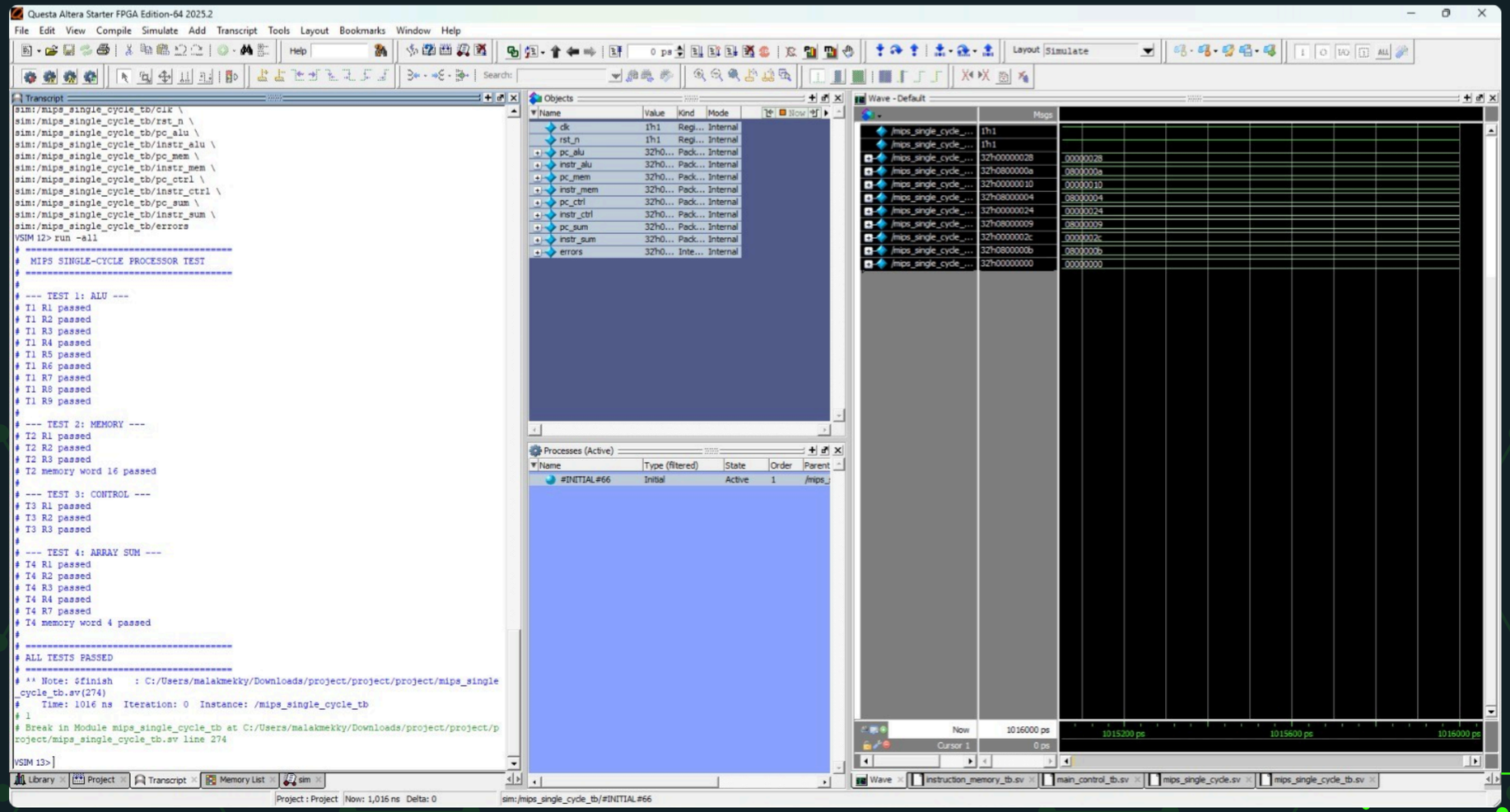
instruction memory TB (2/2)



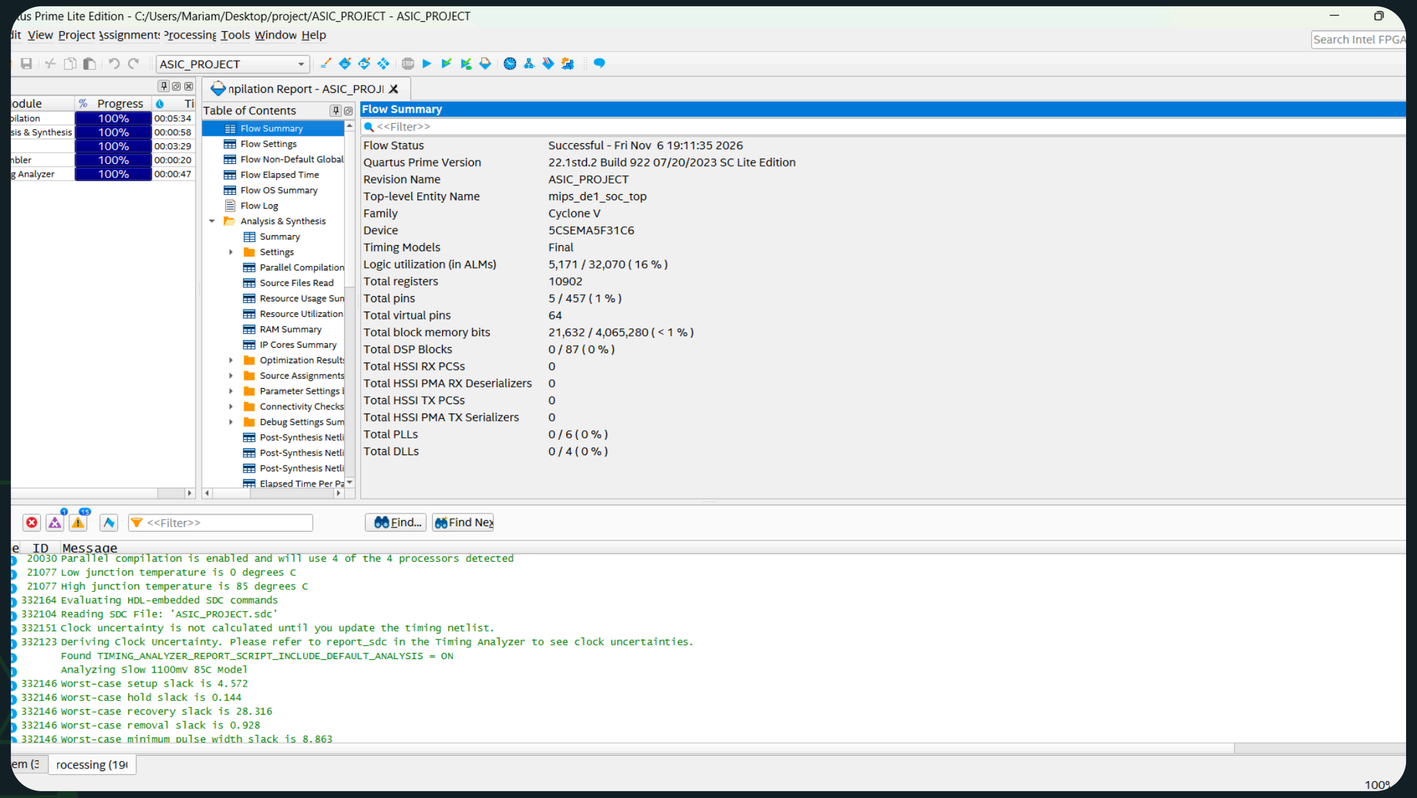
next pc logic TB



mips single cycle TB

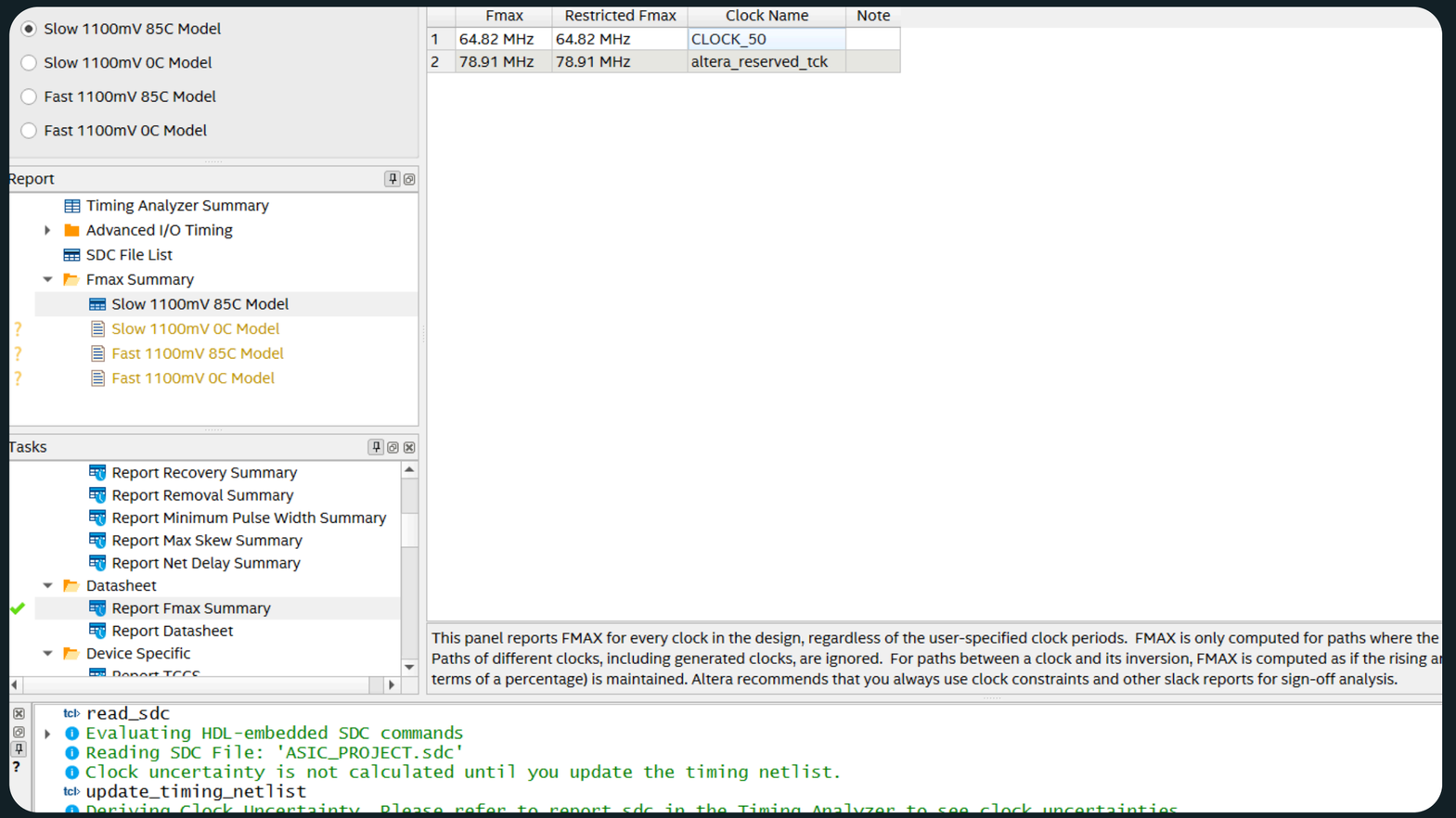


FPGA Resource Usage



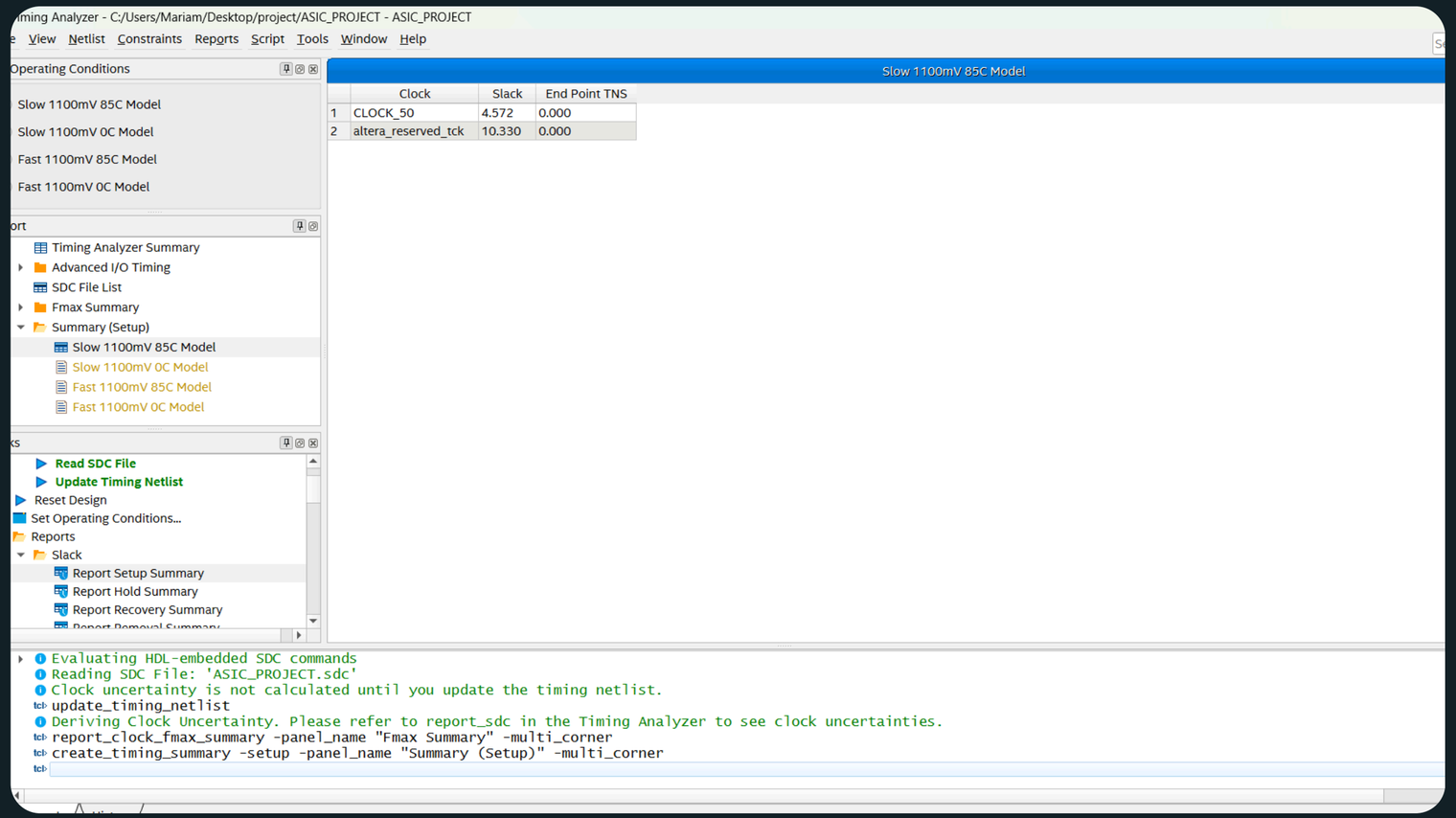
- 5,171 / 32,070 ALMs (16%)
- 10,902 registers
- 21,632 memory bits in 5 M10K blocks, all used by SignalTap
- 0 DSP blocks
- Data memory = 8,192 flip-flops. clock-free read can't use M10K RAM

Timing Analysis: Fmax



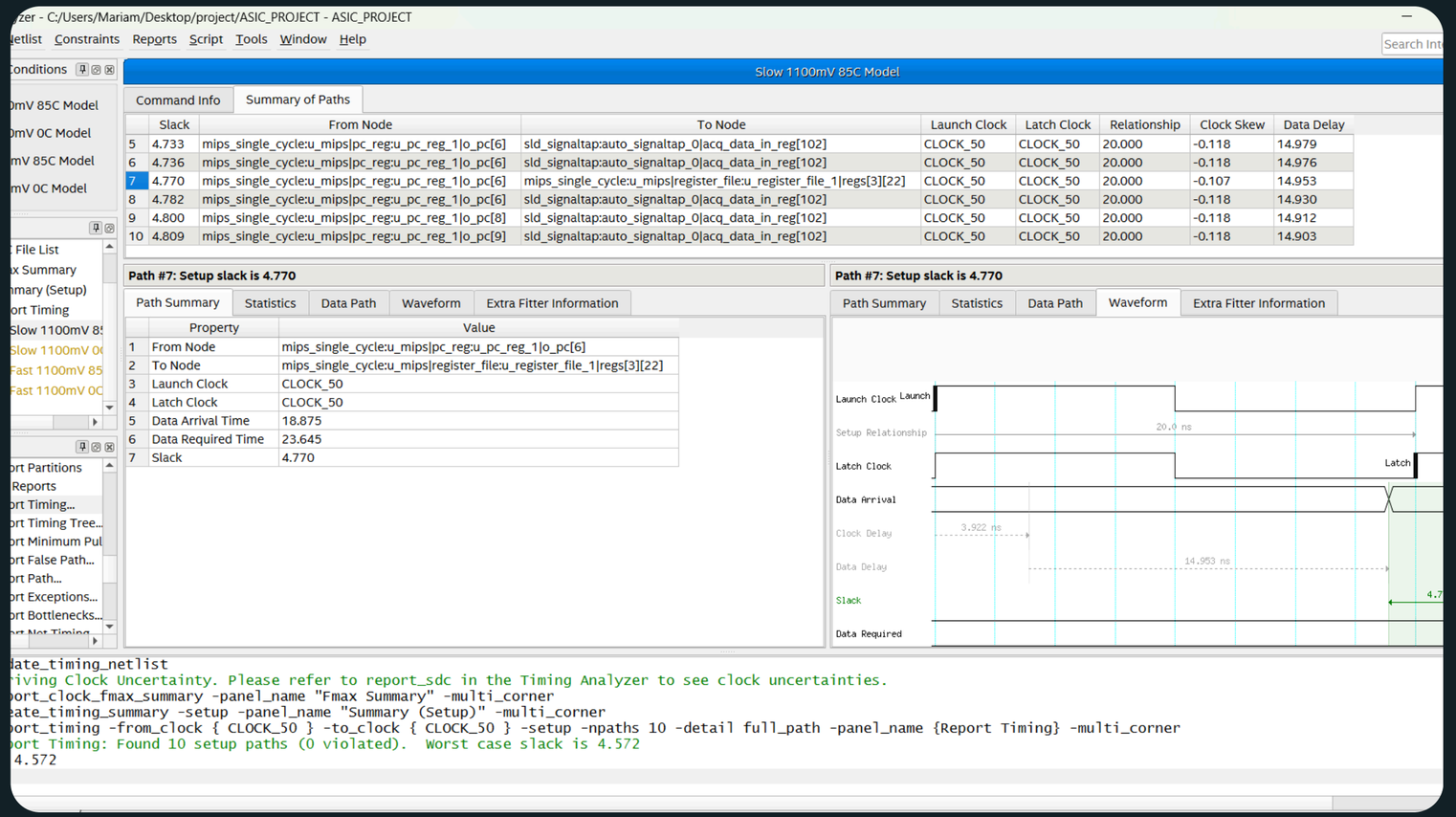
Fmax = 64.82 MHz

Timing Analysis: Worst Slack



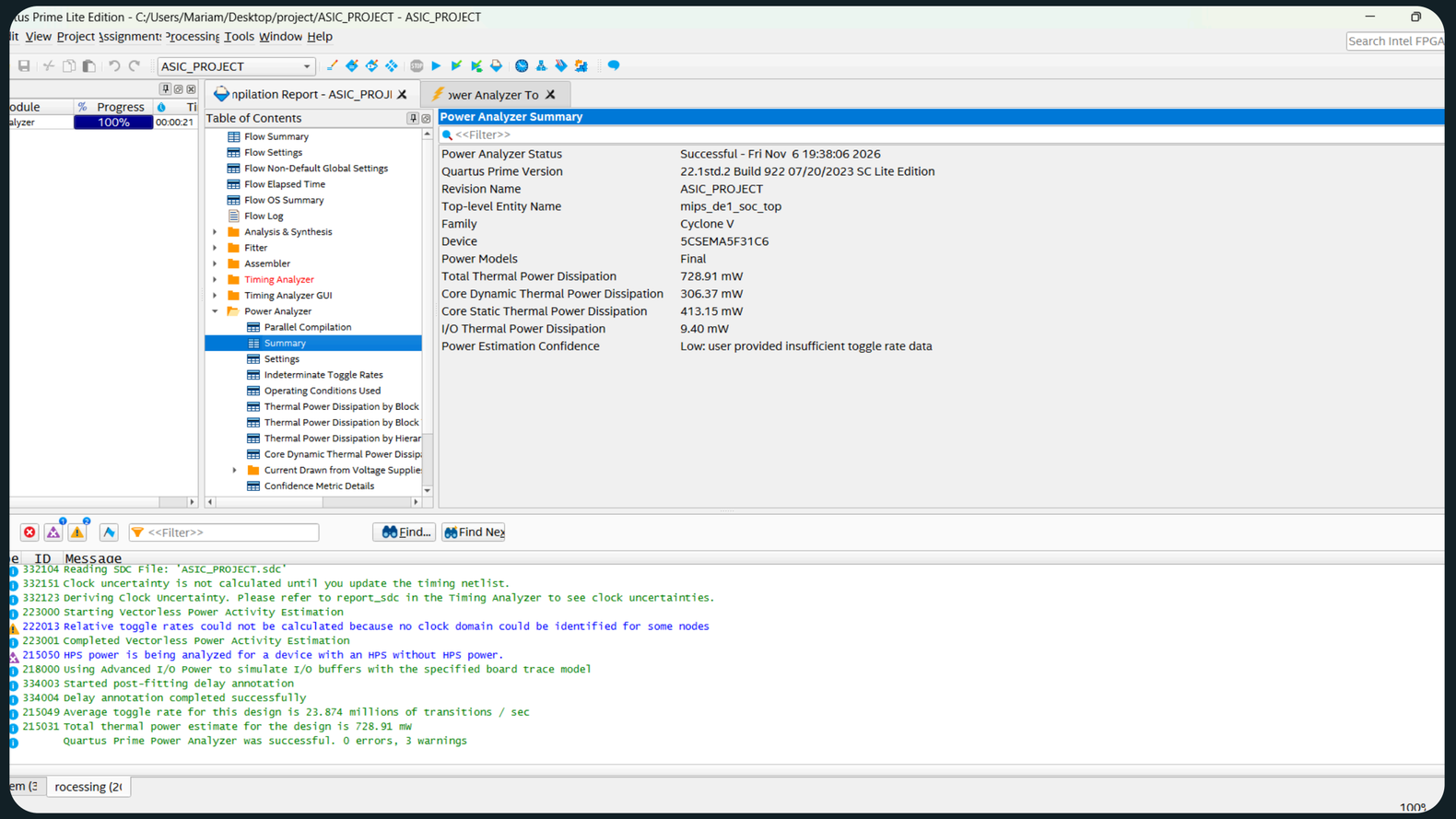
Worst setup slack 4.572 ns at 50 MHz

Timing Analysis: Critical Path



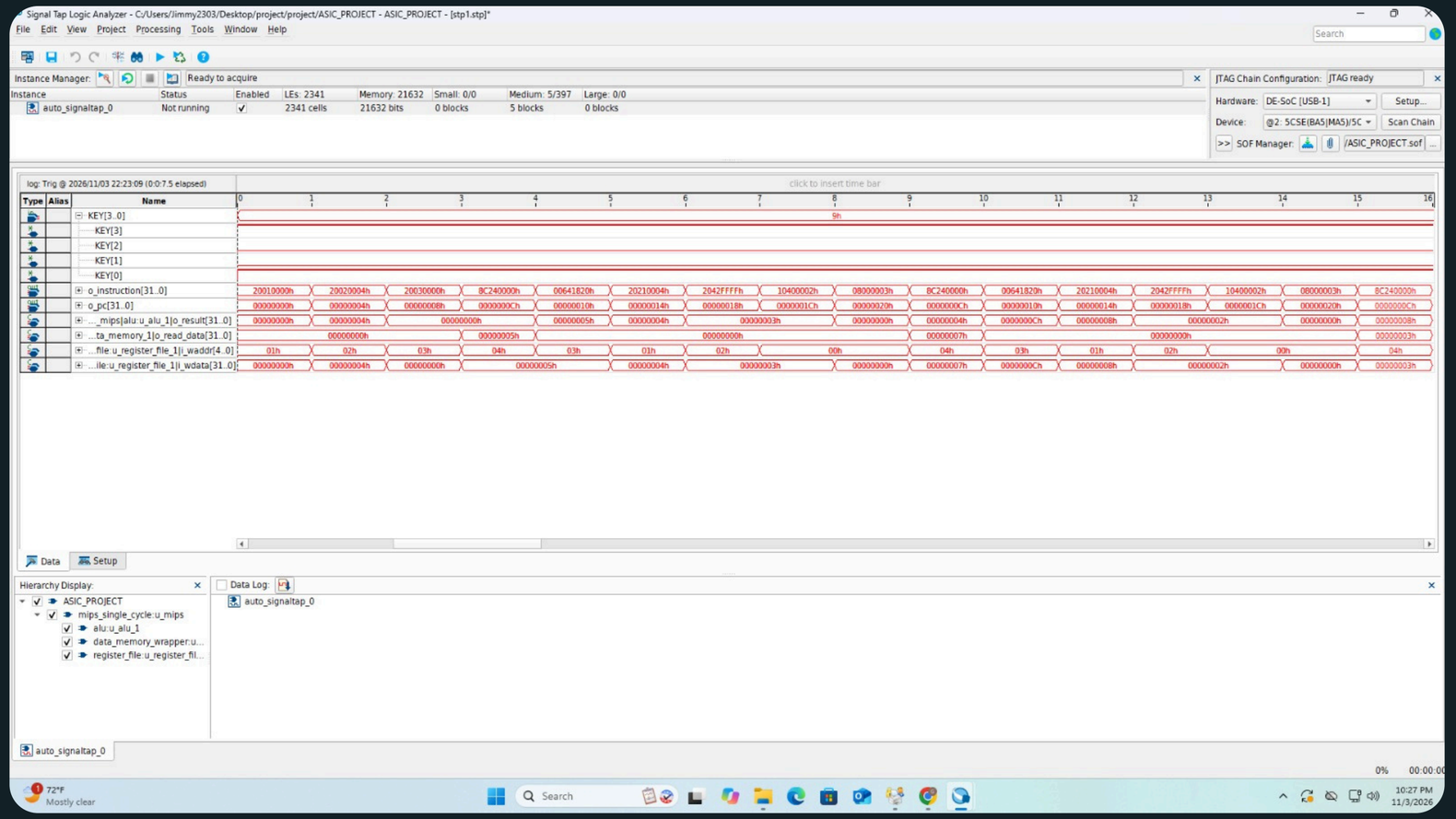
Critical path: PC → instr. memory → register file → ALU → register write (15 ns)

Power Analysis



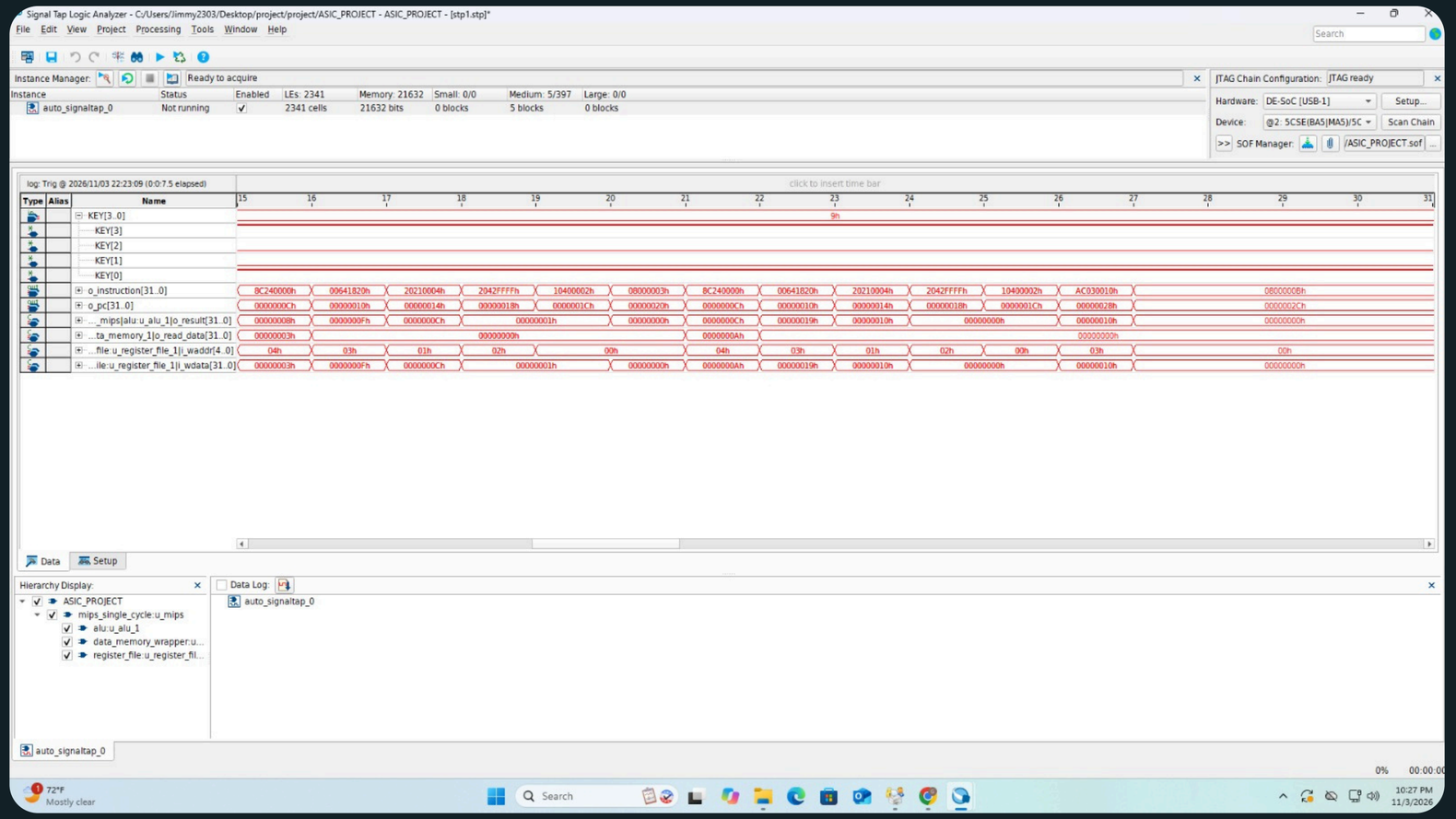
Total Power: 728.91 mW. Static: 413.15 mW. Dynamic: 306.37 mW.

Board Demo



Test 4 Array Sum (1/2)

Board Demo



Test 4 Array Sum (2/2)

Problems & Limitations

Problems

- Register file TB left the new reset unconnected
- No clock constraint, so no real Fmax or power
- Array Sum Test 4
- Warning We Left as is: 6 input / 57 output ports unconstrained

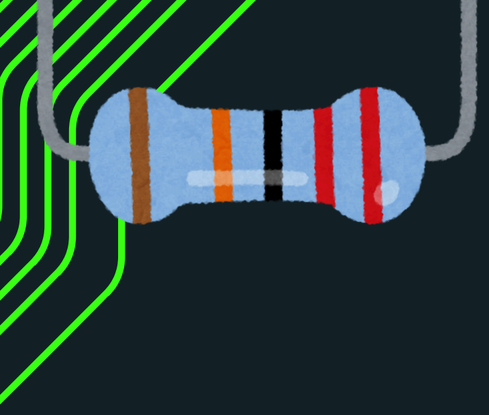
How We Fixed Them

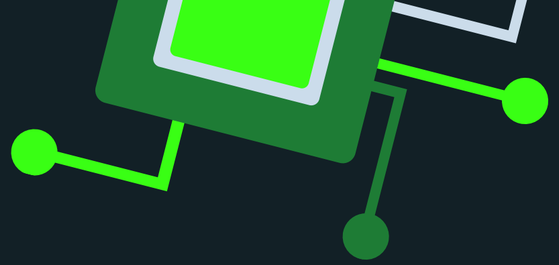
- Connected .rst(rst) in the TB
- Added ASIC_PROJECT.sdc with a 50 MHz clock
- Added reset to register file so we can have initially 0

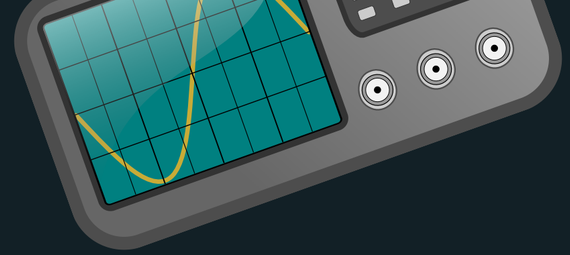
Limitations

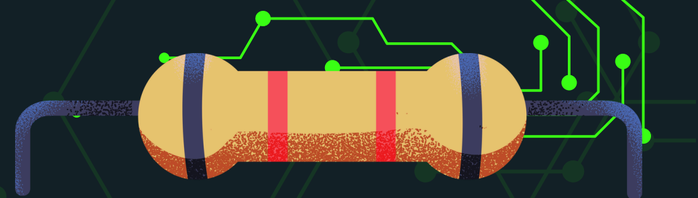
- Data memory uses 8,192 flip-flops instead of RAM (combinational read)
- Power is a vectorless, low-confidence estimate
- FPGA numbers depend on the program in the ROM

Questions?









Thank
You!
GPU